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Review Article

A review on carbon-based phase change materials for thermal energy storage

Raghvendra Kumar Mishra [a], Kartikey Verma [b,*], Vinayak Mishra [c], Babulal Chaudhary [d]

[a] IMDEA Materials Institute, Madrid 28906, Spain
[b] Department of Chemical Engineering, Indian Institute of Technology Kanpur, Kalyanpur, Kanpur, Uttar Pradesh 208016, India
[c] International Institute of Information Technology, Hyderabad 500032, India
[d] Indo-US Science and Technology Forum, New Delhi 10001, India

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ABSTRACT

Phase change materials are attractive as well as being selected as one of the incredibly fascinating materials relating to the high-energy storage system. Phase change materials (PCM) can absorb as well as release thermal energy throughout the melting and freezing process. In recent times, the phase change materials are the feasible option for lowering down the energy consumption and for enhancing the thermal comfort in houses. In this paper, a comprehensive summary of the phase-change materials (PCM) in state of the art is presented. Primary objective of this review is to overview most of the open scientific studies on regarding fatty acid, and carbon allotropes based fatty acid, fatty alcohol phase change materials. Also, this paper is based on earlier scientific studies to consider the variety of references that have been discussed by the authors as well as viewed as the leading contribution.

1. Introduction

Continuously rising greenhouse gas emissions and raising the cost of fossil fuels, the application of renewable power sources and improved energy efficient method has turned out to be more and more vital in the nowadays [1,2]. The thermal energy storage system is necessary for the effective utilisation of renewable energy, and it likewise helps to enhance the energy efficiency of a variety of applications as well as techniques. Thermal energy storage has been employed as an illustration to retain heat received by solar energy [3] throughout sunlight as well as release it throughout the night [4]. Another essential application of thermal energy storage is basically in space heating systems to enhance the energy efficiency of buildings because a possible solution for the conservation of energy to complete the gap between energy source as well as need, through phase change materials have gained consideration significantly [5]. Phase change materials (PCMs) are the type of materials, which absorb or discharge a considerable amount of heat throughout the phase change at a particular temperature [6]. Thermal energy storage with PCM is a favourable technology based upon the concept of latent heat [7]. The stored or discharged heat is referred as the latent heat, where phase change material absorbs or discharges a substantial amount of energy at a specific temperature throughout the phase change transition interval, with an excellent heat of fusion near the phase change temperature range [8], as mentioned in Fig. 1.

The use of phase change material (PCM) is being formulated in a variety of areas such as heating as well as cooling of household, refrigerators [9], solar energy plants [10], photovoltaic electricity generations [11], solar drying devices [12], waste heat recovery as well as hot water systems for household [13]. The two primary requirements for phase change materials are an appropriate phase change temperature for a particular application along with their latent heat [14]. There are a few other specifications, these types of specifications can be classified into thermal, physical, kinetic, chemical as well as economic characteristics for example thermal properties (substantial specific heat, latent heat, thermal conductivity), chemical characteristics (Not having any flammability, long-term chemical stability, absolutely no corrosivity, have zero toxicity), kinetic properties (lesser supercooling, considerable crystallization rate), physical properties (high density, nominal vapour pressure, congruent melting, minimal volume transformation on phase change), and so on [5]. Some phase change materials must be accessible at low-cost. In practice, different types of phase change materials exhibit different specifications. For the examples, the majority of the phase change materials possess poor thermal conductivity, and particularly salt hydrates exhibit considerable super cooling in the time of

* Corresponding author.
E-mail address: vrmakrtk@gmail.com (K. Verma).

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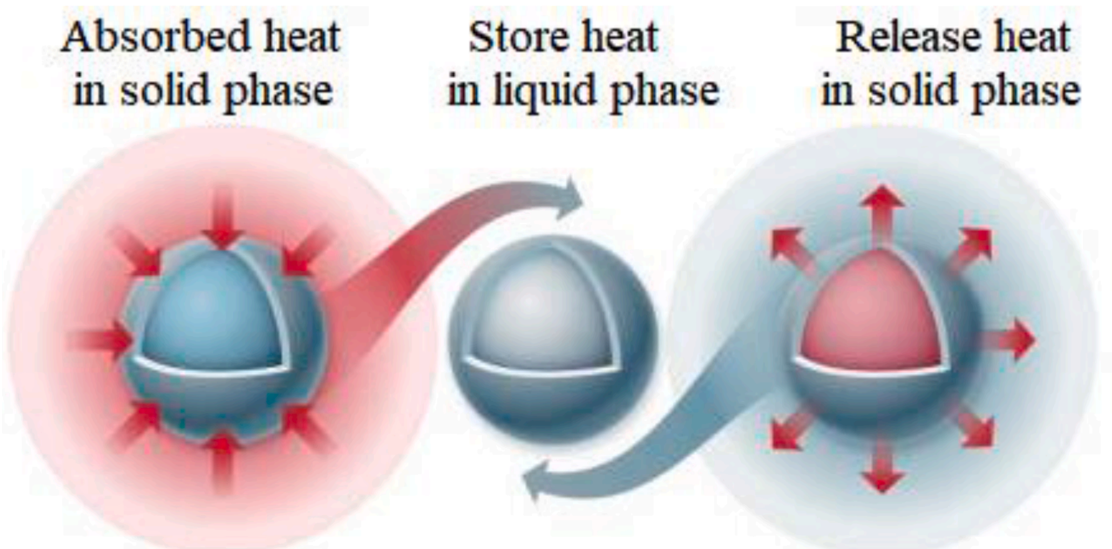

Fig.. 1. Working Principle of Phase Change Materials.

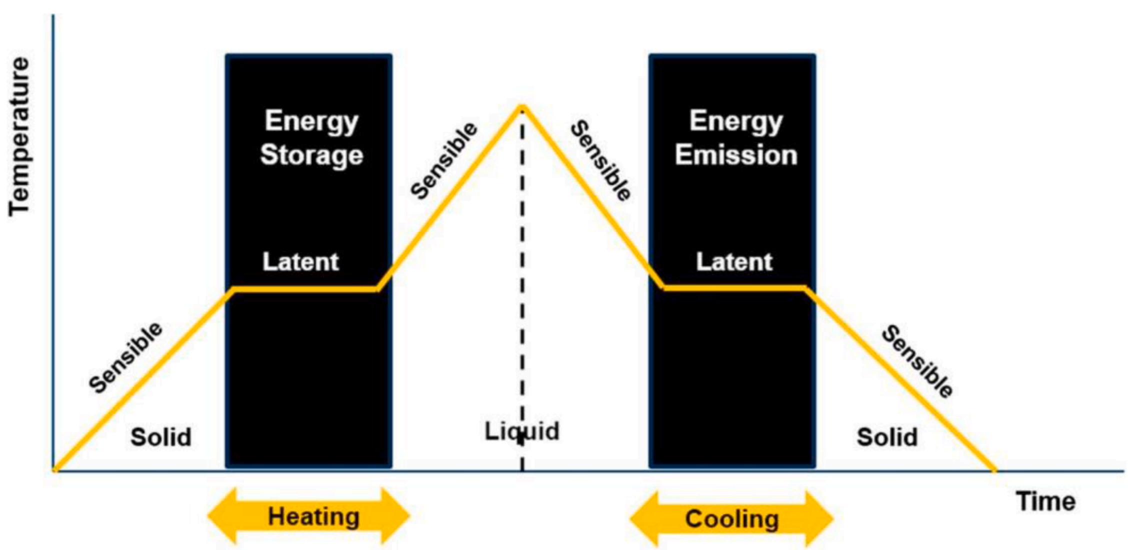

Fig.. 2. Schematic diagram of the phase change transition for Phase Change Materials [18].

crystallisation [15]. In the matter of solid-liquid phase change, additionally, describe "heat of fusion" can be applied. PCMs with solid-liquid phase transition are generally desired due to their significant latent heat as well as small change in volume (generally lower than 10%) throughout the phase change [16]. The liquid-vapour phase change of materials generally possesses significant latent heat. Inconveniently, high vaporisation temperatures, as well as substantial volume change in the course of vaporisation, render the exploitation of liquid-vapour phase change unrealistic behaviour [17]. Thermal energy storage system with PCMs is a viable system according to the theory of latent heat and thermal energy storage, as demonstrated in Fig. 2. This paper covers the state of art of phase change materials (PCMs) based on fatty acid, alcohol, and carbon allotropes.

2. State of the art of phase-change materials

The depleting stocks of fossil fuel, the escalating greenhouse gas emissions, along with the swift evolution of energy ingestion have shed light on the significance of efficiently utilising energy [19]. Therefore, the growth and development of brand-new options for energy is the aim of various scientific studies. The application of PCM for thermal energy storage offers an appropriate solution, inexpensive together with economical energy storage, for enhancing the functionality of energy systems thereby lowering peak requirement and energy intake. From the research of Alan Tower Waterman of Yale University in the early 1900s, breakthrough of phase change material begins [20]. To store thermal energy, Phase change materials (PCMs) or latent heat storage materials are one of effective applied strategies. Energy per unit mass is preserved throughout phase changes from solid to liquid and also discharged throughout freezing at a steady temperature [21]. Material absorbs the energy, and it enables improving the vibrational energy states of the atoms or molecules. Throughout melting, the atomic bonds loosen along with the material alterations its state from solid to liquid are happened [22]. In the course of solidification, the material exchanges energy and, accordingly, the molecules eliminate energy and order themselves in the solid state.

Consequently, latent heat storage systems are small cost and more straightforward to work in comparison with those thermo-chemical heat storage systems. Thermal energy storage contains latent heat storage as well as thermo-chemical heat storage. Due to its low energy density,

Table. 1
Nanotechnology in Phase change materials.

Type of Phase Change Materials	Types of fillers	Remarks	References
Water-based nanofluid	Al_2O_3 nanoparticle	A considerable effect of nanoparticle focuses on thermal properties of phase change materials; nanoparticle decreases total charging time.	[33]
De-ionised water (DI) based	Multi-walled Carbon Nanotubes	MWCNT act as a nucleating agent provides the energy saving potential	[34]
Bio-based composite	Silica fume and exfoliated graphite nanoplatelets	A heightening in the thermal conductivity of the Bio-based PCM is achieved by the silica fume and exfoliated graphite nanoplatelets.	[35]
Pure water	Nanoscale Cu particles	Cu nanoparticles help to boost the thermal conductivity of the water-based coolant.	[36]
Water	Al_2O_3	By using Al2O3, the friction factor is amplified. However, long-term stability, agglomeration is found to be a significant problem.	[37]
Paraffin (n-octadecane)	emulsifying alumina (Al_2O_3) nanoparticles	The alumina (Al2O3) nanoparticles improve viscosity, the thermal conductivity of paraffin.	[38]
Polyurethane	Carbon nanotube sponge	Aligned carbon nanotubes (CNTs) network helped to improve the energy conversion/transfer performance.	[39]
Polyurethane	Graphene oxide	Phase change behaviour has been altered concerning the size of GO.	[40]
Soy wax and paraffin wax	carbon nanofiber (CNF) and carbon nanotube (CNT)	Increasing the CNF or CNT loading enhances the thermal conductivity of the PCM despite this, CNF is more excellent filler than CNT for the thermal conductivity.	[41]
Paraffin	Graphite nanofibers	The thermal effect of graphite fibre is dependant on the type of fibre such as herringbone, ribbon, or platelet.	[42]
Organic ester	Silver–titania hybrid	Improvement in viscosity, thermal stability, heat storage is found concerning the loading of the silver–titania hybrid.	[43]
capric acid/ stearic acid/ palmitic acid mixture	Nano-silicon dioxide	Nano-silicon dioxide helps to achieve the high latent heat value, excellent chemical and thermal stability, the low thermal conductivity for the stabilise the indoor temperature for a long duration as well as achieve desired human comfort, throughout 500 heating/cooling cycles.	[43]
Paraffin	Multi-walled carbon nanotubes (MWCNT)	The diameter of multi-wall carbon nanotube can affect the latent heat, in this way, smaller diameter MWCNT leads to attaining lower latent heat of the nanofluid. Smaller diameter can enhance the thermal conductivity.	[44]
Octadecanoic acid	Three-dimensional graphene aerogel	Three-dimensional graphene aerogel helps to achieve the excellent thermal conductivity and high heat storage capacity.	[45]
Water	Titanium dioxide nanoparticles	The phase transition temperature, phase transition time as well as enthalpy of nanofluid are altered concerning the content of TiO_2 at the low cooling rate.	[46]
Stearic acid	Al_2O_3 coated by graphene	Graphene-coated Al_2O_3 improves the thermal conductivity, wettability and reduces the interfacial thermal resistance of the stearic acid.	[47]
1-octadecanol (stearyl alcohol)	Graphene	In comparison to the other nanomaterials such as carbon nanotubes, silver nanomaterial, Graphene significantly increases the thermal conductivity of nanocomposites. However, the reduction in the heat of fusion has been found.	[48]
Paraffin	Nanomagnetite	The inclusion of nano magnetite has improved storage capacity, heat transfer properties, Thermal conductivity of the paraffin.	[49]
Tetradecanol	copper nanowires	The doping of the copper nanowire has improved the thermal conductivity. While the copper nanowires have decreased phase change enthalpy.	[50]
Palmitic acid	Nitrogen-doped graphene	The addition of Nitrogen-doped graphene has increased the thermal conductivity, electrical conductivity, thermal reliability and chemical durability of nanocomposites.	[51]
Palmitic acid/ Polyvinyl butyral	Expanded graphite	Expanded graphite helps to increase thermal conductivity as well as reduce the leakage problem of PCMs.	[52]
Paraffin wax	Hybrid graphene aerogel	The thermal conductivity, high thermal energy storage density, excellent thermal reliability and chemical stability of paraffin wax have been achieved by hybrid graphene aerogel.	[53]

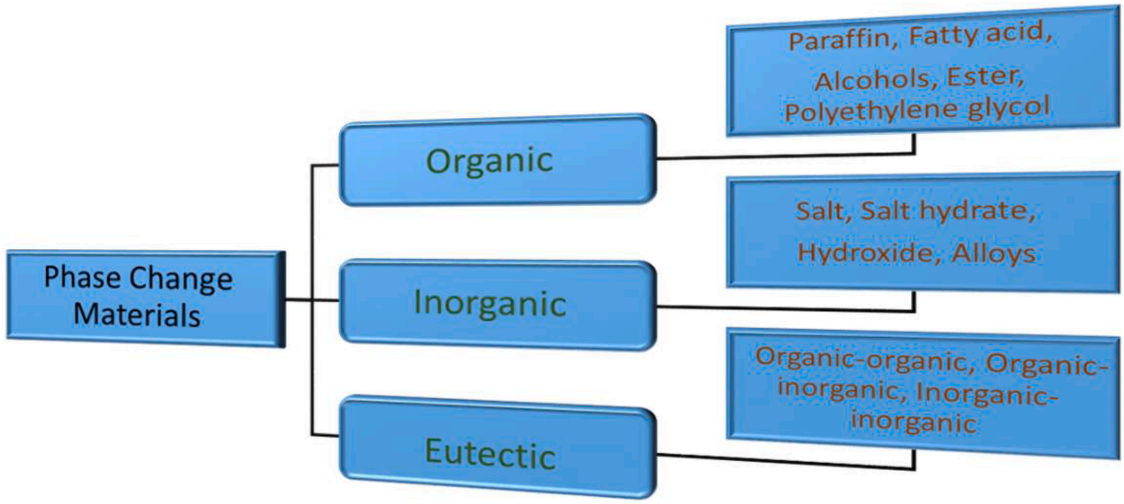

Fig.. 3. Classification of Phase change materials.

sensible heat storage demands large volumes as well as entails appropriate design to release thermal energy [23]. Within this background, phase change materials -based cooling system is an extremely appealing strategy of thermal control. Taking into account the benefits of the phase change materials, high latent heat storage ability, smaller volume alternation in the phase transition, superior specific heat capacity to render extra sensible heat storage [24]. In addition to chemical stability, merely no degradation for a sheer number of cycles, phase transformation takes place inside the preferred working temperature range of the system. Higher nucleation rate helps to prevent super-cooling of the liquid phase. Sufficient thermal conductivity leads to help the absorption and discharge of energy in the storage system, reasonably inexpensive, and excessive accessibility [25]. Alternatively, various options have been designed to boost the heat transfer in PCMs by the help of metal oxide nanofillers [26], graphite [27], carbon nanotubes [28]or fabric [29], and exfoliated graphite [30], carbon nanofiber [31], graphene [32], as mentioned in Table. 1.

3. Types and properties of phase change materials

There are many forms of phase change materials available. Primarily, there are three forms: Organic (paraffin as well as nonparaffin), Inorganic (salt hydrates and also metallic alloys), and Eutectic (combination of two or more-phase change materials elements: organic, inorganic, and even both), as shown in Fig. 3 [54]. In the past years, the scientists concentrate on phase change material wherein heat storage is performed because of latent heat of fusion [55].

In addition to the latent heat of fusion numerous organic, as well as inorganic materials, are applied in the role of phase change material primarily based on the melting point and freezing point. Their melting and freezing point are available in the operating range, most of the phase change materials cannot fulfil the criteria for thermal energy storage device because no single material may have almost all the attributes necessary for thermal energy storage [31]. Consequently, the materials have to be employed and their thermos-physical attributes should be enhanced by creating appropriate adjustments in systems design or by applying exterior agents [54]. Wherein organic phase change material hold various attributes for example substantial specific heat, broad and changeable melting point range, accessible in colossal temperature range, low vapour pressure in melt manner, heat of fusion, tiny volume changes throughout phase transition, chemical as well as thermal stability, non-corrosives, recyclable, minimal or probably none under cooling [7,56]. Paraffin wax, n-octadecane, and so on are the example of organic phase change materials such as. It shows an extensive temperature range and also much better chemical stability [57].

There are many disadvantages with organic phase change material, such as low thermal conductivity, relative substantial volume change, non-compatible with plastic containers, flammability, lower phase change enthalpy [58]. Aside from organic phase change materials, inorganic phase materials also provide some outstanding attributes, for example, acute phase change, lower environmental effect, non-flammable, better phase change enthalpy, excellent thermal conductivity, and also substantial latent heat of fusion [8]. Inorganic PCMs such as salt hydrates, alternatively, possess better thermal conductivity and lower volume change however they cool down swiftly and corrosive. Nevertheless, there is undoubtedly a disadvantage with inorganic phase change materials, for example, substantial volume change, inadequate nucleation rates, higher vapour pressure (induce water loss as well as result in progressive change in thermal characteristic throughout thermal cycling method), long-lasting degradation by oxidation, hydrolysis, thermal decomposition as well as other reactions, corrosion as well as irritant, under cooling etc. [59]. Eutectics possess the excellent thermal energy storage capacity with regards to the volume as well as accessible for very high-temperature applications also. Eutectic provides several benefits, for example, sharp melting temperature above organic compounds, no segregation and congruent phase change, volumetric thermal storage density, and also disadvantageous are constrained information can be found on their thermo-physical attributes [60]. Based upon the phase transformation, phase change materials can be found in the form of solid-solid, solid-liquid, and solid-gas and liquid-gas phase change material [61]. The phase transformations are functioned in both ways, with energy possibly discharged to or absorbed from the surroundings. The energy is thermodynamically symbolised by an alteration of pressure, temperature or volume. Generally, the temperature difference offers the primary effect on a phase change. Probably, the present investigation ideas are to find and characterise new materials which can be used as a suitable phase change material for a precise application. The thermal energy storage area is now a modern technology for saving energy and raising energy efficiency by dealing the deviation between energy supply along demand. Thermal energy storage entails storage capacity of high- alternatively low-temperature thermal energy in the role of latent heat, sensible heat, or perhaps via thermo-chemical reactions as well as processes [62].

Sensible heat storage of a system is dependant on storing thermal energy using cooling or heating a liquid or solid system. Sensible heat storage system involves substantial volumes due to its low energy density. Moreover, sensible heat storage systems have to have a suitable design to release thermal energy at constant temperatures [63]. The heat

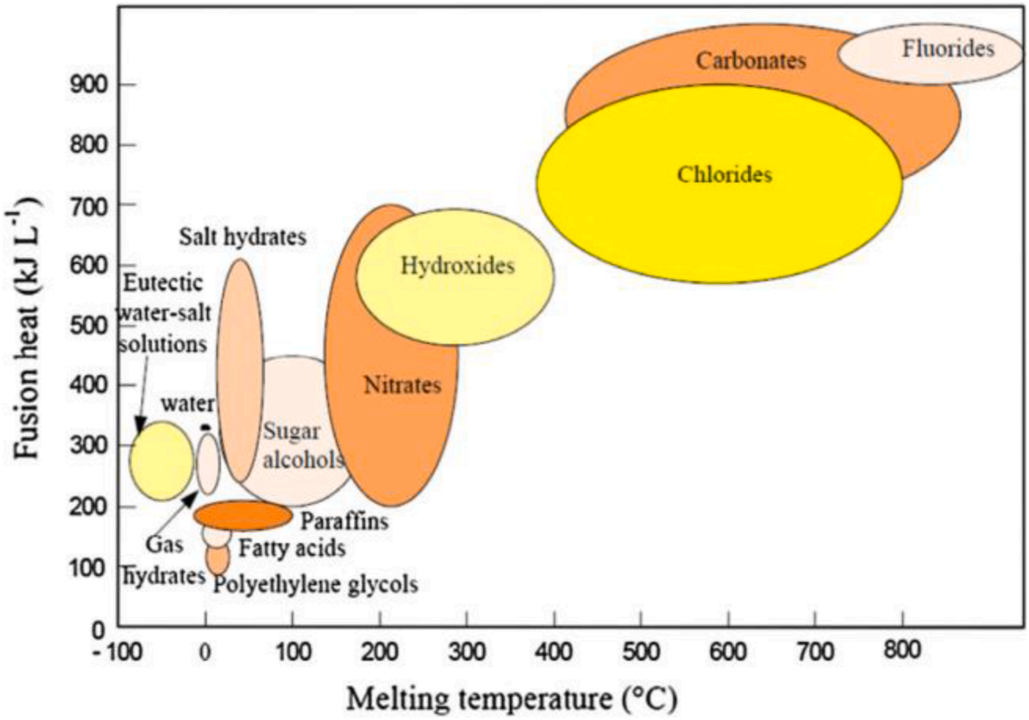

Fig.. 4. Different types of phase change material and their thermal properties [69].

of fusion also referred as enthalpy of fusion describes the quantity of thermal energy that material is required to absorb or discharge as a way to transform its phase from solid to liquid or even vice-versa [64]. The heat of fusion straightly influences storage ability of the phase change materials since strong values of heat of fusion result in more effective systems. High heat of fusion is involved in passive heating/cooling applications for buildings to improve the energy stored, minimising the points of heating/cooling requirement [65]. The primary attributes of phase change materials are the temperature along with the heat of fusion. Generally, it is crucial to containing the temperature of fusion throughout the temperature-based application; however, material's energy storage capacity is not influenced by temperature of fusion [66]. Nevertheless, a phase change is associated with melting; the addition of the temperature of fusion in the temperature range of practical application can allow the function of change in phase in the form of an on-off switch [54]. Fig. 4 outline common tendencies of organic as well as inorganic phase change materials about specific heat and enthalpy, melting point and range [67]. From a thermal perspective, an appropriate phase change temperature range, excellent latent heat of fusion along with a good heat transfer in the direction of the phase change materials are preferred. The latent heat storage ability of phase change materials is inevitably related to the packing density of molecules inside the crystalline system, although the phase transformation temperature is dependant upon the strength of the non-covalent bonding between molecules [68].

From a physical point of view, phase equilibrium, zero phase segregation, a sufficient density along with small volume changes at the phase change are generally desired for simple application in present house building materials as well as structures [70]. From a kinetic point of view, no additional super-cooling along with an adequate crystallisation rate is usually required to generate the best application of the features as well as probabilities of phase change materials [71]. Super-cooling, the strategy of minimising the temperature of a liquid underneath its freezing point without any changing into a solid, can affect the performance of the phase change materials based on the suitable phase change temperature by influencing this temperature [72]. From a chemical perspective, durable chemical stability of the phase change material despite cycling, compatibility with building materials, non-toxicity without any fire hazard is required [73]. To be employed for applications, the phase change materials a/re needed to have certain appropriate thermos-physical, kinetic, chemical, economic as well as environmentally friendly features. Extraordinary thermal conductivity to assist in charging as well as discharging of phase change materials inside the constrained time frame [71]. Excellent specific heat so that extra energy in the form of sensible heat is found to the thermal energy storage system [74]. Phase change material must melt entirely in the course of a phase transition which means the solid, as well as liquid phases, are homogenous [54]. Thermally efficient in order that phase change material is consistent on phase change temperature as well as latent heat of fusion which enables it to be considered in long-term. It is chemically suitable for construction/encapsulated materials [75,76]. The phase change materials which transform from the solid to the liquid state holds a disadvantage: their leakage in the melting condition is found [77]. Leakage decreases the heat transfer efficiency as well as raises manufacturing expenditure. Therefore, many studies concentrate on microencapsulation, which contains covering the phase change material together with a material which does not connect with the matrix or phase change material. An additional option considered is to include the phase change material straight into the material of the matrix, without having the coating material, in order that the matrix can function as the capsule of the phase change material [78]. A different research group has performed characterisation of phase change material in the form of a thermal management system. The thermal test device which mainly emphasised their use in the area of power electronics, and also authors applied differential scanning calorimetry as well as temperature history technique for application in the role of thermal regulators in photovoltaic systems [79].

4. Organic phase-change materials

Several research studies have highlighted organic phase change materials. The accessible, as well as widely explored organic phase change materials that experience solid-liquid phase transformation in the passage of cooling and heating, are poly (ethylene glycol), paraffin waxes, fatty acids and their derivatives [80]. Additionally, several organic phase change materials, for example, polyalcohols [81] and polyethylenes [82], which have solid-solid phase transformation at a set temperature by absorbing and discharging massive volume of the latent heat, have been gained interest in the form of the favourable organic phase change materials [83]. A variety of requirements must be satisfied by a suitable organic phase change materials. These requirements are substantial latent heat ability to offer an excellent thermal storage density, little volume modification throughout phase transformation, thermal stability during the passage of several heating and cooling cycles, repeatability of phase change, non-flammable, chemically stable, non-toxic, non-corrosive, inexpensive in cost, and readily accessible [8,84,85]. Based on several researches, the drawbacks are the low thermal conductivity held by several organic phase change materials resulting in low charging as well as releasing rates, supercooling consequence in cooling cycles, and the requirement for containers to avoid the leakage of phase change materials [86]. Presently, several research organisations have extensively investigated the thermal properties of paraffin waxes for the growth and progress of various heat storage materials. This research indicates that paraffin waxes store, absorb and discharge plentiful heat repeatedly throughout phase conversions between solid as well as liquid phases. They have significantly latent heat holding capacities between 200 kJ kg^{-1} and 250 kJ kg^{-1}, as well as have a broad range of melting temperatures with thermal stability up to 250 °C. They show no phase segregation in the course of repeated phase transitions. They are noncorrosive, chemically inert, odourless, durable, affordable, readily available, nontoxic and ecologically safe. Due to these excellent features, paraffin waxes and their blends are widely renowned for many industrial thermal storage purposes [87].

Poly(ethylene glycol)s also called poly(oxyethylene), are made up of linear dimethyl ether chains with hydroxyl ending groups, $HOCH_2(-CH_2OCH_2)nCH_2OH$, and consequently, have twin characteristic of water solubility as well as organic solubility. Poly(ethylene glycol)s have already been researched as phase change materials in numerous different thermal storage applications from building envelopes to textiles, foams as well as fibres due to their suitable characteristics. These characteristics are high heat of fusion, low and reasonable melting temperature, low vapour pressure while melted, thermally as well as chemically stable, biodegradable, nonflammable, non-corrosive, non-toxic and also low-priced [88]. It is viable to change the melting temperature and capacity to heat absorption of a poly (ethylene glycol)s system by choosing or blending poly (ethylene glycol)s with various molecular weight. There is a substantial supercooling effect because of the issues of crystallisation of long phase change material chains in the course of cooling [89]. In this manner, avoiding the supercooling of poly (ethylene glycol) s apart from the enhancement of their thermal conductivity is a significant concern for their near future application in heat storage systems. Several scientists have examined the fatty acids as phase change materials. Mainly, in the current scientific studies, the fatty acids are a widespread concern in the phase change materials area. As the fatty acids possess several outstanding characteristics, for example, appropriate melting temperature range and also high heat capacity, the super-cooling effect are ignored throughout the transformation, with decent chemical and thermal stability, less expensive, or little volume change [90].

Additionally, the thermal conductivity is enhanced by introducing graphite, which enhances the rate of thermal storage and discharge. Depending on the features, it is useful to employ them to the solar energy storage, thermal storage system, as well as other fields. Furthermore, the fatty acid esters have a lesser phase change temperature. Therefore it is more efficiently employed in the solar-powered air-conditioning system [91].

4.1. Fatty acids and fatty acid derivatives

Notably, the fatty acid (FA) is a carboxylic acid having a very long aliphatic chain, that is possibly saturated or unsaturated. The majority of natural fatty acids possess an unbranched chain of an even number of carbon atoms, ranging from four to twenty-eight. Fatty acids are generally found in three primary categories of esters: triglycerides, phospholipids, and cholesterol esters. For such varieties, fatty acids are vital nutritional options for fuel for animals, so they are essential structural parts for cells [92]. Fatty acids are different in length, generally classified in the form of short to very long. Short-chain fatty acids have aliphatic tails of five or maybe very few carbons. Medium-chain fatty acids are fatty acids having aliphatic tails of six to twelve carbons, that may shape medium-chain triglycerides. Long-chain fatty acids are fatty acids that have aliphatic tails of thirteen to twenty-one carbons. Long chain fatty acids are fatty acids that have aliphatic tails of twenty-two or higher carbons. Saturated fatty acids do not comprise of a C = C double bond [93]. They have already the same formulation $CH_3(CH_2)$ nCOOH, accompanying distinctions in "n". An essential saturated fatty acid is stearic acid ($n = 16$), which while neutralised using lye is one of typical kind of soap [94]. Some of Saturated Fatty Acids are Caprylic acid, Lauric acid, Capric acid, Palmitic acid, Myristic acid, Stearic acid, Behenic acid, Arachidic acid, Cerotic acid, Lignoceric acid,. Caprylic acid is the general term for the eight-carbon saturated fatty acid recognised by the regular designation octanoic acid [95]. Its compounds are seen in the milk of a variety of mammals, as well as a minor component of coconut oil and also palm kernel oil. It is an oily liquid which is merely soluble in water having a little undesirable rancid-like smell as well as flavour [96,97]. Decanoic acid also referred to as capric acid or decylic acid is a saturated fatty acid. Its chemical formula is $CH_3(CH_2)8COOH$. Salts, as well as esters of decanoic acid, are known as decanoates or caprates. The terminology capric acid is obtained from the Latin "caper / Capra" due to the fact the sweaty, undesirable fragrance of the compound is an indication of goats [98]. Attention in the application concerning fats as well as oils for eco-friendly chemistry resulted in the increasing significance regarding materials made from them. The animal in addition to plant-centred fats and oils are usually hydrolysed to have combinations of fatty acids which can be filtered and afterwards isolated. Presently, fatty acids in the form of a phase change materials receive much consideration for their several features in building energy performance, air-conditioning systems as well as solar heating systems. Fatty acids grow to be amidst sustainable feedstocks, with attributes equivalent to paraffin waxes as part of phase change material concepts. The fatty acids and their various types of derivatives have now been considered in the role of an appealing phase change materials to receive energy-retaining compounds in solar energy systems, construction, as a result of their relatively effectual thermal, physical properties as well as their simple introduction into a composite [99]. Additionally, the majority of the fatty acids are usually on the market, because some companies currently create fatty acids in enormous quantities for plastic materials, beauty products, fabric Additionally other companies. Around 1989, D. Feldman has evaluated the thermal characteristics of lauric, palmitic, and also capric stearic acids and their binary blends [100]. The statistics confirmed that these types of items are alluring applicants for latent heat retaining as part of space heating reasons. The melting range of these types of fatty acids changed ranging from 30 °C to 65 °C, although their enthalpy of fusion values have been found in between 153 kJ kg−1 to 182 kJ kg^{-1}. D. Feldman et al. performed a substantial study concerning the application of fatty acids in the role of a phase change materials also their thermal stabilities [101]. A. Hasan analysed the thermal characteristics of stearic acid, also palmitic acid [102]. A. Sari also analysed the thermal abilities of myristic acid [103]. F.O. Cedeno et al. introduced the melting temperature also the heat of fusion description data for palmitic, stearic also oleic acids also their binary also ternary blends [104]. G. Gbabode observed the thermal characteristics also the crystal elements of odd-numbered fatty acids ($C_nH_{2n+1}COOH$) from tridecanoic acid ($C_{12}H_{25}COOH$) to tricosanoic acid ($C_{22}H_{45}COOH$) [105]. In this manner, we have outlined the properties of fatty acid-based phase change

Table. 2
Fatty acid based phase change materials.

Type of organic Phase Change Material	Density (kg/m^3)	Thermal conductivity (W/mK)	References
Elaieimic acid	901 in liquid phase, 981 in solid phase	0.149 in liquid phase	[106]
Myristic acid (MA)	990 in solid phase, 861 in the liquid phase	0.150 in the liquid phase	[107]
Glyceric acid	850 in the liquid phase, 989 in solid phase	0.162 in the liquid phase	[92]
Stearic acid (SA)	965 in solid phase, 848 in the liquid phase	0.172 in the -liquid phase	[108]
Paraffins	790 in the liquid phase, 916 in solid phase	0.167 in the liquid phase 0.346 in solid phase	[109]
Palmitic acid (PA)	989 in solid phase, 850 in the liquid phase	0.162 in the liquid phase	[110]
Capric acid (CA)	1004 in solid phase, 878 in the liquid phase	0.153 in the liquid phase	[111]
Lauric acid (LA)	1007 in solid phase, 862 in the liquid phase	0.150 in the liquid phase	[112]

materials in Table 2.

Consequently, several experts concentrated on the thermal characteristics of the fatty acids, their blends also their forms for evaluating their suitability for heat retaining functions. Fatty acids possess specific unwanted characteristics, such as inferior smell, corrosivity, and particularly elevated sublimation rate, to conquer the drawbacks, a variety of fatty acid forms are typically made by esterification reactions of fatty acids with the help of alcohol. Fatty acid esters are commonly a type of organic phase change material, and inadequate thermal data is in the literature [113]. Nowadays, studies have been concentrated on low-chain fatty acid esters of stearic as well as palmitic acids. The melting point of these types of unique materials occur between 20 °C to 40 °C, and related heat capacities range from 180 kJ kg^{-1} to 200 kJ kg^{-1}, correspondingly [114]. Authors carried out the DSC characterisation to study the melting process of stearic acid, oleic acid, palmitic acid and their binary, ternary blends. Due to the fact oleic can go through two-phase transformations (solid–solid, solid-liquid) throughout the temperature-heighten and temperature-fall stage, the melting steps of the binary as well as ternary blends experienced a couple of endothermic peaks [115]. Authors examined the thermal features of stearic acids during the freezing as well as the melting process. The final results reveal that the emerging fin improves both the conduction as well as similarly the natural convection heat transfer of phase change material, and the improvement factor throughout freeze has been calculated, it was 250% higher. The experimental results reveal that the improvement mechanism of the fin is linked to its potential to enhance both heat conduction and also natural convection [116]. There are several studies on the phase change materials based on the fatty acid; we have outlined a few of those study in Table 3.

4.1.1. Fatty acid composites as a phase change material

Fatty acids are significantly used in thermal energy storage because there are countless benefits to create fatty acids into stable phases change materials, such as without necessity for further encapsulation, stable shape, cost-advantages, and the ability for uses along with tunable dimension. Fatty acids in the role of a phase change materials posses little thermal conductivity, which retards the storage as well as heat discharge rate during the melting and solidification steps, and low thermal conductivity of organic PCMs obstructs their further uses in thermal energy storage [92,131]. By applying the various fillers,

Table. 3
An overview of Fatty acid as a Phase change material.

Fatty acid-based Change Material	Remarks	References
A mixture of capric and lauric acids	It is useful for low-temperature storage.	[117]
lauric acid(LA), palmitic acid(PA) and their Binary systems	Provide the details about solid-liquid phase transitions at various temperatures, lauric– the palmitic acid binary system shows thermal stability after 100 heating-cooling cycles at 32.8 °C, it can be used in the dynamic energy storage.	[118]
The eutectic mixtures of myristic and stearic acids	Low-temperature thermal storage applications	[119]
Esters of myristic, palmitic and stearic acids with glycerol	Melting temperatures of these have been found in the range between 31 °C(corresponding latent heat 149 kJ kg^{-1}) and 63 °C (corresponding latent heat 185 kJ kg^{-1}). These esters based phase change material offers excellent thermal stability even after 1000 thermal cycles.	[120]
Capric acid(CA)–Lauric acid(LA) /PMMA form-stable phase change material	Phase change temperatures and latent heat values are 21.11 °C and 76.3 kJ/kg respectively, application in building energy conservation.	[121]
Capric acid– myristic acid /PMMA form-stable phase change material	Phase change temperatures and latent heat values are 25.16 °C and 69.32 kJ/kg respectively, application in building energy conservation.	[122]
Capric acid– stearic acid /PMMA form-stable phase change material	Phase change temperatures and latent heat values are 26.38 °C and 59.29 kJ/kg respectively, application in building energy conservation.	[122]
Lauric acid – Myristic acid /PMMA form-stable phase change material	Phase change temperatures and latent heat values are 34.81 °C and 80.75 kJ/kg respectively, application in building energy conservation.	[122]
Stearic-capric acid- activated-attapulgite stable phase change material	The phase change temperature and latent heat are 21.8 °C and 72.6 J/g respectively. Thermal and chemical stable concerning a large number of thermal cycling. Application in latent heat thermal energy storage for the building energy conservation.	[123]
Capric acid-palmitic acid-stearic acid ternary eutectic mixture/ nano-silicon dioxide stable phase change material	The phase change temperature has been found in the range from 17.16 °C–26 °C, latent heat value and thermal conductivity are 99.43 kJ kg−1 and 0.08239 W (m K)−1. Application for storing solar radiation and minimising the heat loss of buildings in the winter.	[124]
Shape-stabilised lauric acid/ activated carbon-based phase change material	Latent heat is enhanced with increasing the content of the lauric acid content in composites, and thermal conductivity is improved. The phase change behaviour of the composite is the same as lauric acid. Activated carbon leads to enhance the thermal stability of the composites.	[125]
Eutectic mixtures of lauric acid–myristic acid	A melting temperature of 34.2 °C, latent heat of fusion of 166.8 J $g-1$, excellent thermal properties and thermal reliability during 1460 thermal cycles	[126]

(continued on next page)

Table. 3 (*continued*)

Fatty acid-based Change Material	Remarks	References
Eutectic mixtures of lauric acid–palmitic acid	A melting temperature of 35.2 °C, latent heat of fusion of 166.3 J *g* − 1, excellent thermal properties and thermal reliability during 1460 thermal cycles	[127]
Eutectic mixtures of myristic acid–stearic acid	A melting temperature of 44.1 °C, latent heat of fusion of 182.4 J *g* − 1, excellent thermal properties and thermal reliability during 1460 thermal cycles	[127]
MA/HDPE/ Nano–Al_2O_3 composites	Thermal conductivities of Nanocomposites based phase change material is increased by95% at 12% nano-Al2O3 content. Application in thermal energy storage	[128]
MA/HDPE/ nano–graphite composites	Thermal conductivities of Nanocomposites based phase change material is increased by 121 at 12% nano-graphite content. Application in thermal energy storage.	[129]
Palmitic acid/ high density polyethylene/ graphene nanoplatelets Nanocomposite	Thermal conductivity, as well as thermal stability, is improved by graphene nanoplatelets, the graphene nanoplatelets significantly reduce leakage of palmitic acid.	[130]

significant number of researches had been done previously to improve the heat transfer efficiency of fatty acid and increase the thermal energy absorbing/releasing speed [132]. The presently applied materials having outstanding heat conduction feature consists of carbon fibre, expanded graphite (EG), carbon nanotubes (CNTs), graphene, activated carbon, silica fume (SF) and activated montmorillonite (a-MMT), expanded perlite [133,134] and so on. These fillers are used to generate the fatty acid composite. Generally, there are primarily two factors for establishing fatty acids composites in the form of a phase change materials. First one is to promote the heat conduction potential of phase change materials because fatty acids in the role of a phase change materials display low heat conduction possibility; Enhancing thermal conductivity is just one of the primary objective in the area of phase change materials. The mentioned scientific studies in current years concerning thermal conductivity improvement of PCMs through incorporating fillers with high thermal conductivity encapsulating phase change materials, wherein the fillers primarily consist of carbon-based materials as well as metal-based materials. Fig. 5 reveals the strategies of thermal conductivity improvement.

Moreover, the positive aspects and drawbacks of every technique for enhancing the thermal conductivity are described as well as reviewed, the objective of offering the reader with a comparatively systematic and comprehensive understanding of all of them [135]. Moreover, another factor is to understand the stable system as well as solidification of phase change materials. There are numerous techniques to prepare and stabilise phase change materials such as microencapsulation, adsorption, electrostatic spinning, polymer blends. We cover the fatty acid composite phase change material based on the carbon allotropes. Therefore, a review of the fatty acid and carbon-based composite in the role of a phase change material is mentioned in the following section.

4.1.1.1. Carbon based phase change materials. Energy storage is important area in saving energy as well as providing the safeguard to the environment. The phase change materials play a vital role in the forefront of the thermal storage field. Although, the poor thermal conductivity and the loss during phase transition hinders the application range of the phase change materials (PCMs). To solve this problem, Carbon-based materials play a crucial role, and it has broad range of the application in the field of phase of materials. Carbon materials in PCMs is used to enhance thermal conductivity, mechanical, electrical and adsorption properties. In this section, applications of carbon nanotubes, carbon fibers and graphite, graphene in fatty acid based have been discussed.

4.1.1.2. Graphite based fatty acid phase change material. Earlier studies

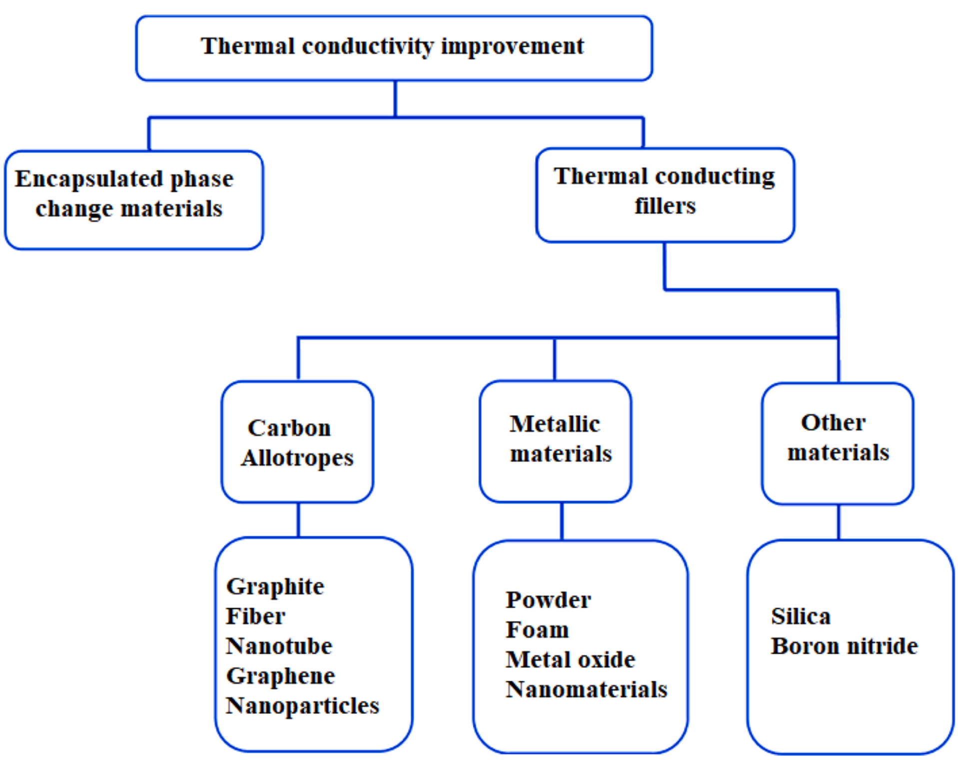

Fig.. 5. The strategy of thermal conductivity enhancement.

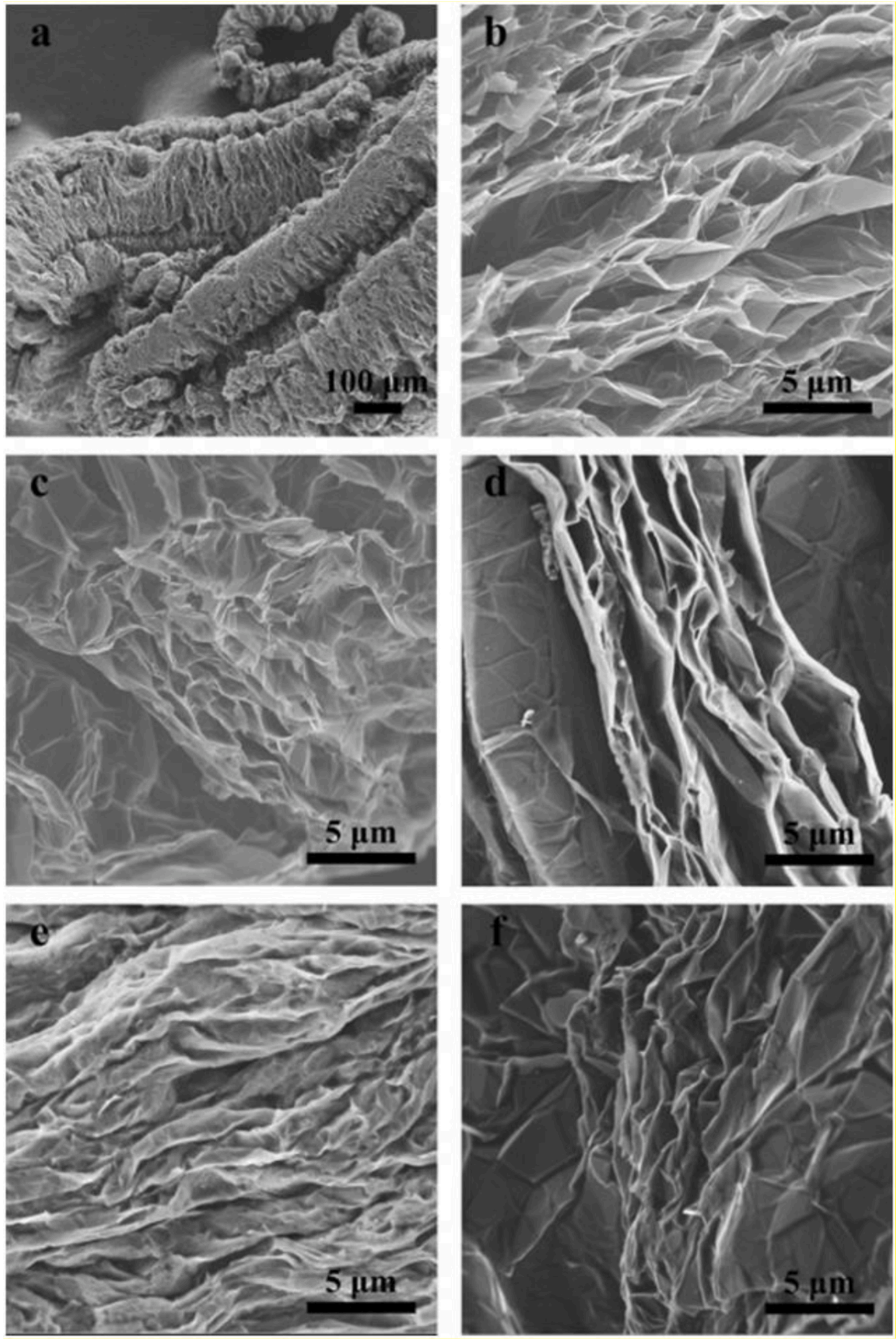

Fig.. 6. SEM images of (a and b) raw expanded graphite, (c) SPCMs 1, (d) SPCMs 3, (e) SPCMs 5, and (f) SPCMs 7[143].

have proven the fact that the thermal conductivity of PCMs can be enhanced by using expanded graphite. As a result, the thermal conductivity improvement of expanded graphite based organic phase change materials has drawn research preferences. Expanded graphite having a porous structure is additionally desired as its excellent thermal conductivity, reasonable density, and suitable compatibility along with organic PCMs. Authors mentioned that the thermal conductivity of paraffin/6 wt% expanded graphite composites had been improved from 0.32 to 3.66 W/m K in comparison to paraffin [136]. Authors blended acetamide with expanded graphite, therefore, improving the thermal conductivity from has been reported [137]. Authors described that the thermal conductivity of stearic acid had been enhanced by expanded graphite. Authors described that the thermal conductivity of lauric acid (LA), myristic acid(MA), and palmitic acid(PA) had been boosted by expanded graphite [138]. Expanded graphite along with a perfectly-formed pore structure can decrease the seepage of liquid phase change materials by using capillary force. Furthermore, the pore structure can restrict the comparative significant volume alteration of the phase change materials depending on fatty acid eutectic throughout their phase change steps [139,140]. Capric acid, lauric acid, and oleic acid have been blended in a certain percentage to produce a eutectic mixture, accompanied by absorbing into the expanded graphite to create the composite shape-stabilised phase change materials [141,142].

SEM illustrations have been applied to study the microstructures of the expanded graphite and the as-prepared composite phase change materials, as shown in Fig. 8. It has been found from Fig. 6a that the expanded graphite displays as the worm-like microstructure. Furthermore, a huge level of properly-derived pores network at the microscale

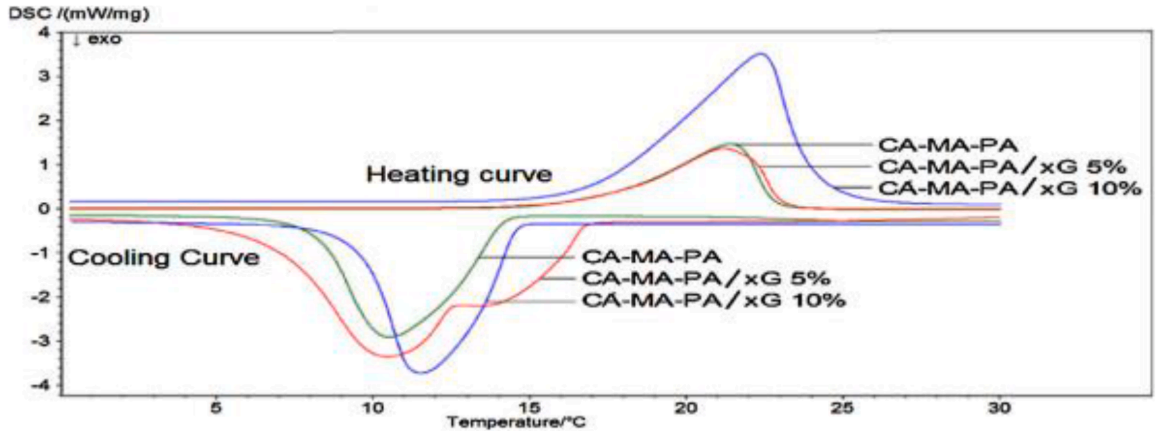

Fig.. 7. DSC Curve of CA–MA–PA and CA–MA–PA/xG composite PCM [146].

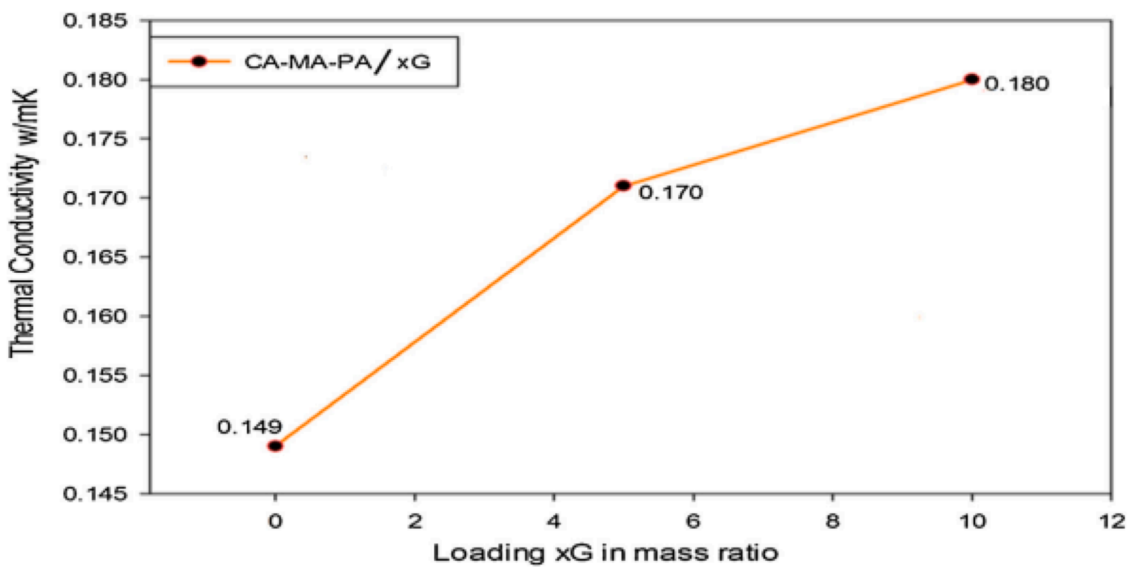

Fig.. 8. The thermal conductivity of CA–MA–PA and CA–MA–PA PCM composites loaded with xG[147].

in worm-like expanded graphite has been precisely noticed (Fig. 6b), suggesting that the expanded graphite presents a considerable surface area. Therefore, it favours good absorption capability. Conversely, as a result of the capillary force of the porous microstructure in expanded graphite, the impact of the volume alteration of C-L-O acid throughout its solid-liquid transition process has been sidestepped. Fig. 6b additionally shows the soft surface and layer like the edge of the expanded graphite. When, C-L-O acid has been absorbed into the micropores of the raw expanded graphite (Fig. 6c–f), the layer like edges and surface turned out to be rougher, and the layers become thicker [143].

Fatty acid and expanded graphite composites in the form of a phase change materials for latent heat thermal energy storage systems have been developed through vacuum impregnation technique. The capric, lauric, and myristic acids in the role of a phase change material have been applied due to the fact of their excellent properties. The expanded graphite have been employed in the role of an adsorbent material because of to its features of the porous structure, excellent thermal conductivity, industrial diversity [143]. Stearic acid-expanded graphite composites having various mass proportions have been developed through absorbing liquid stearic acid onto the expanded graphite. The stearic acid in composite materials, has been applied in the form of a phase change material for application in thermal energy storage, and the expanded graphite served for the supporting material. The Stearic acid-expanded graphite composites solidify at 54.28oC which have a latent heat of 155.70 kJ/kg, and also Stearic acid-expanded graphite composites have been melted at a temperature of 53.12oC having a latent heat of 155.50 kJ/kg the moment the mass proportion of the stearic acid in the composites is up to 83% [143]. Lauric acid-myristic acid-expanded graphite composite phase change materials (PCMs) have been developed as a result of absorbing the binary eutectic mixtures into expanded graphite (EG). The thermal stability of Lauric acid-myristic acid-expanded composite PCMs magnifies in contrast to Lauric acid-myristic acid eutectic mixtures [144].

A ternary eutectic blend of capric–myristic–palmitic acid and exfoliated graphite (capric acid–myristic acid–palmitic acid/exfoliated graphite) composites in the form of a phase change material (PCM) for energy storage has been reported. Fig. 7 demonstrates the DSC curves of capric acid–myristic acid–palmitic acid and capric acid–myristic acid–palmitic acid/exfoliated graphite composite phase change material. The melting temperature, as well as freezing temperature of capric acid–myristic acid–palmitic acid/exfoliated graphite composite, had been found to be relatively lower than that of capric acid–myristic acid–palmitic acid. Because the exfoliated graphite has a more excellent thermal conductivity which escalates the heat transfer rate and it helps to decrease the phase change temperature. It has been reported that the melting as well as freezing latent heat values of capric acid–myristic acid–palmitic acid/exfoliated graphite have become considerably lower than capric–myristic–palmitic acid [145].

Table. 4
An overview of Graphite based Fatty phase change materials.

Graphite and Fatty acid based Change Material	Remarks	References
Stearic Acid–Acetanilide Eutectic Mixture/Expanded Graphite composite	High thermal conductivity, large heat storage capacity, and exceptional thermal stability and promising latent heat energy storage.	[148]
Stearic acid/graphite sheets composite	Good thermal stability below 180 °C, composites has excellent thermal reliability and thermal conductivity of 10.08 times higher than Stearic acid. Ideal to use in building thermal energy storage system and solar thermal storage.	[149]
Hollow porous Co_3O_4 on expanded graphite/ stearic acid composite	Superior thermal performances than stearic acid block the leakage of molten stearic acid.	[150]
Stearyl alcohol/ palmitic and stearic acid/ graphite	Thermal conductivity increased by a factor of six, Single melting and crystallisation peaks observed.	[151]
Palmitic acid/polyvinyl butyral/ expanded graphite	Palmitic acids have a higher latent heat, thermal conductivity of palmitic acid/ polyvinyl butyral composites can be remarkably improved by expanded graphite.	[152]
Palmitic/ stearic acid/ expanded graphite	Optimum absorption ratio, improved in thermal conductivity, boosted thermal energy storage and release rates, excellent thermal reliability and stability.	[153]
Capric-palmitic-stearic acid/ expanded graphite composite	Increasing thermal conductivity, reduction in melting/cooling time, excellent thermal reliability after 500 heating/cooling cycles test.	[154]
Palmitic acid(PA)/polyaniline/ exfoliated graphite nanoplatelets (GNP)	Improvement in the thermal conductivity, lowest thermal energy storage capacity, shows better performance in solar thermal energy applications.	[155]
Myristic acid/graphite nanoplates composite	Thermal cycling test, repeated melting/freezing cycles, Improved in thermal conductivity of the composite.	[156]
Fatty acid eutectics/expanded graphite	Thermal conductivity improvement of low melting point, enhancement in thermal conductivity, excellent in thermal reliability.	[143]
Myristic–palmitic–stearic acid/ expanded graphite	Proper phase change temperature, high thermal conductivity and latent heat, stability for thermal energy storage in solar heating, excellent thermal reliability, waste heating recovery systems and other potential applications.	[157]
Palmitic acid/polyvinyl butyral/ expanded graphite composites	Improvement in thermal energy storage systems, especially in low-temperature solar energy systems, Expanded graphite enhances thermal conductivity, help to reduce leakage.	[152]
Palmitic acid/polyvinyl butyral/ expanded graphite composites	Improvement in latent heat and thermal conductivity, thermal energy storage systems, especially in low-temperature solar energy systems.	[152]
Fatty acid eutectics/bentonite/ expanded graphite composites	Improving thermal properties, excellent thermal stability after 50 heating–cooling cycling, enhancement in thermal storage and release rate.	[158]
Stearic acid (SA)/expanded graphite (EG) composite	Enhancement in the thermal performance, enhancement in thermal stability, favourable for low-temperature solar energy storage.	[159]
Lauric–myristic–stearic acid/ expanded graphite	Better thermal reliability after 1000 thermal cycles, applications in energy buildings, condensation heat recovery.	[160]
Capric–myristic–palmitic/ expanded graphite	Good thermal reliability, latent heat storage system, enhancement in the thermal conductivity and energy store rate.	[161]

The thermal conductivity of capric–myristic–palmitic acid, as well as capric acid–myristic acid–palmitic acid/exfoliated graphite composites, have been presented in Fig. 8. It has been noticed that the thermal conductivities capric–myristic–palmitic acid PCM composites has been remarkably enhanced by the inclusion of the exfoliated graphite. The thermal conductivity of capric acid–myristic acid–palmitic acid/exfoliated graphite composites using a mass proportion of 5 wt% offer 0.17 W/mk that is just 14% enhance, however, 10.0 wt% enhanced around 20.8%, and it reveals the result of 0.18 W/mK [147].

Earlier researches have proven that the thermal conductivity of PCMs can be enhanced by the inclusion of expanded graphite. For that reason, the thermal conductivity improvement of expended graphite-based organic phase change materials has drawn numerous research interests. An overview of the previous studies has been mentioned in Table 4.

4.1.1.3. Carbon fibers based fatty acid phase change material (PCM). The poor thermal conductivity issue of fatty acid in the role of a phase change material, some researches have been carried out with the objective of establishing energy storage systems. Carbon fibre (CF) and Carbon fibre brushes having a high thermal conductivity (190–220 W/mK) have been employed to improve the heat transfer in energy storage systems [162]. Authors investigated phase change materials (PCM) based on the carbon for application in thermal energy storage. In this manner, expanded graphite and carbon fibre/stearic acid (SA) phase change materials having various mass proportion and thermal conductivity have been examined. The result suggested that thermal conductivity can be considerably enhanced by introducing expanded graphite (EG) and carbon fibre although the latent heat storage capacity has not been considerably decreased [163]. Thermal conductivity enhancement of stearic acid by expanded graphite and carbon fibre has been reported. Expanded graphite (EG) and carbon fibre in various weight proportions (2%, 4%, 7%, and 10%) have been incorporated to stearic acid, and thermal conductivities of expanded graphite /stearic acid and carbon fibre/stearic acid composites have been calculated. The thermal conductivity of stearic acid (0.30 W/mK) has been enhanced by 266.6% by the inclusion of 10% weight proportions expanded graphite and carbon fibre. The melting times of composite PCMs have been decreased considerably in comparison to stearic acid. It has been decided that the application of expanded graphite and carbon fibre can be seen as a successful strategy to improve the thermal conductivity of stearic acid without decreasing its latent heat storage capacity [123] substantially.

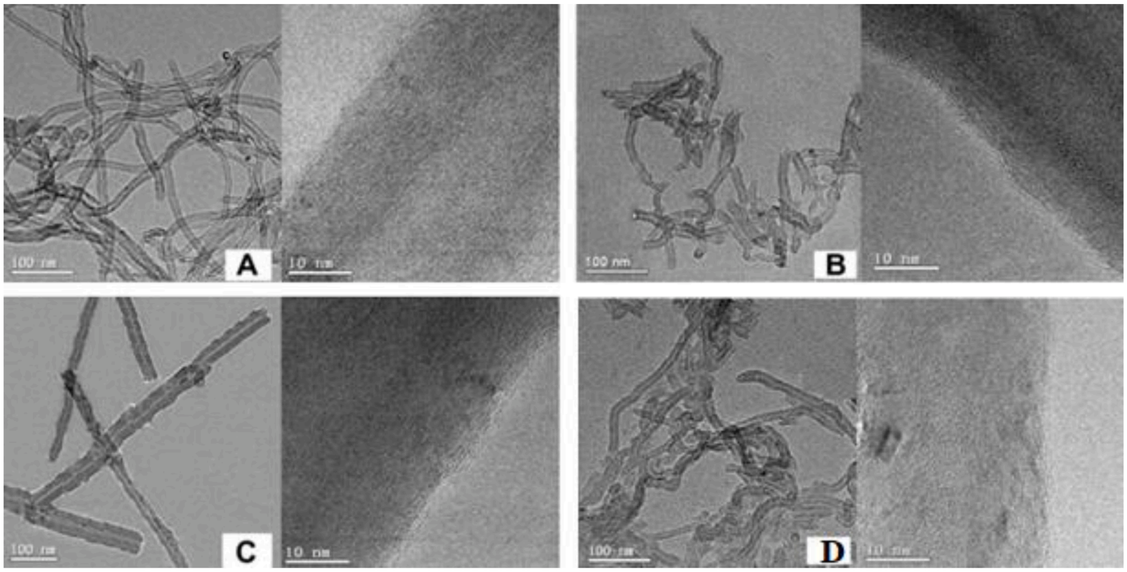

Fig.. 9. Transmission Electron Microscopy [TEM] images of P-MWCNTs (A), B-MWCNTs (B), M-MWCNTs (C), and G-MWCNTs (D)[173].

The effect of expanded graphite as well as carbon fibre as being heat diffusion supporters on thermal conductivity enhancement of stearic acid (has been analysed. The enhancement in thermal conductivity of stearic acid has been additionally examined by evaluating melting time of the pure stearic acid, stearic acid/expanded graphite and stearic acid/carbon fibre composites. It was suggested that the application of expanded graphite and carbon fibre are considered to be a successful strategy to enhance the thermal conductivity of SA without having a reduction in its latent heat storage capacity [164]. Studies based on carbon electrospun mat based fatty acid phase change material has been reported, electrospun nanofibrous mats have been extremely porous as well as able of absorbing a huge quantity of phase change material, and the maximum absorption abilities of carbon mats has been mentioned, it is around 81 wt%. The data suggested that the melting temperatures, as well as enthalpies of the prepared composite PCMs, have been in the range of 18.8–30.7 °C and 92.1– 128.0 kJ/kg for carbon mats. It is expected that this kind of stable PCMs can be applied for building energy conservation [165]. The impact of carbon nanofibers on structural morphologies, thermal conductivity and storage characteristics, of electrospun lauric acid/PA6 composite fibres, has been investigated. the lauric acid/PA6/carbon nanofibers ultrafine composite fibres experienced preferred properties, and it can be used as stable phase change materials [166]. The thermal characteristics of carbon nanofibers (CNFs), as well as carbon nanotubes (CNTs), loaded phase change material has been mentioned to enhance the thermal conductivities [166]. Also, the surface treatment of carbon nanofibers influences the thermal conductivity of the phase change material (PCM). Surface modified-carbon nanofiber based phase change material has shown excellent improvement in thermal conductivity [167].

4.1.1.4. Carbon nanotubes based fatty acid phase change material. As the milestone breakthrough by Iijima in 1991, that was a carbon nanotube. The past decade, carbon nanotubes have drawn considerable attention for applications in the form of fillers in composite materials because of to their fascinating properties such mechanical, thermal, electrical and so on [168]. Composites with CNTs in the form of fillers can improve the mechanical, electrical, as well as thermal properties of the matrix materials [169]. In the situation of thermal transport, carbon nanotubes reveal remarkable capabilities for applications due to the fact of their high thermal conductivity. Authors measured the thermal conductivity of carbon nanotube by application of equilibrium molecular dynamics simulations, and they received a value of thermal conductivity around 6600 W/m K at room temperature [168]. Authors outlined the thermal conductivity of a single-walled CNT, and they mentioned 2980 W/m K at the room temperature. The modified Carbon nanotubes (CNTs) have been incorporated into palmitic acid to formulate composite phase change material. The thermal conductivity enhances of the composite phase change material is highly dependant on the modification of the carbon nanotubes process [170]. Many research has been published in which carbon nanotube have been integrated into the base fluids as well as phase change materials (PCMs) to promote the thermal conductivities. It has also drawn terrific attention on the interfacial heat flow as well as the relationship between the interfacial heat flow and the functionalization of the carbon nanotube [171]. Interfacial thermal resistance in the carbon nanotubes, as well as the matrix, is considered to interfacial heat flow. Authors applied classical molecular dynamics (MD) simulation to analyse the influence of the chemical fictionalisation on thermal transport behaviour of carbon nanotube composites [172]. They noted that poor bonding between various matrix materials with nanotube walls is certainly one of the main factors for the elevated value of interfacial resistance. Chemical bonding between tube and matrix decreases interfacial resistance. Authors observed that percolation performed an important role for tube-tube as well as tube-substrate thermal resistance variables commonly in carbon nanotube composites. No existing theory can separately explain the interesting heat transport phenomena in CNT composites constantly.

Authors developed carbon nanotube composites having palmitic acid (PA) in the role of the matrix. Fig. 9 provides the microstructure of the multi-walled carbon nanotube. Fig. 9A shows that the P-MWCNTs are found in aggregated and entangled morphology. Unmodified P-MWCNTs have not formed stable suspensions in melt palmitic acid. The MWCNT dispersibility has been improved by chemical and mechanical treatments because chemical or mechanical treatment support to disrupt the P-MWCNT aggregates. Ball milling process reduced the aspect ratio of the MWCNT significantly, as shown in Fig. 9B and 9C. Fig. 9D shows that additional attachments on the surfaces of G-MWCNTs have been reported. The modification of MWCNT has played a significant role to modify the MWCNT surface, which performed an important part to enhance the thermal conductivity of the MWCNT/ palmitic acid composites [173].

Palmitic acid-stearic acid eutectic mixture has been developed. Subsequently, a develop of palmitic acid-stearic acid/carbon nanotubes (CNTs) composite phase change materials (PCMs) have been formulated to increase the thermal conductivity of the phase change materials

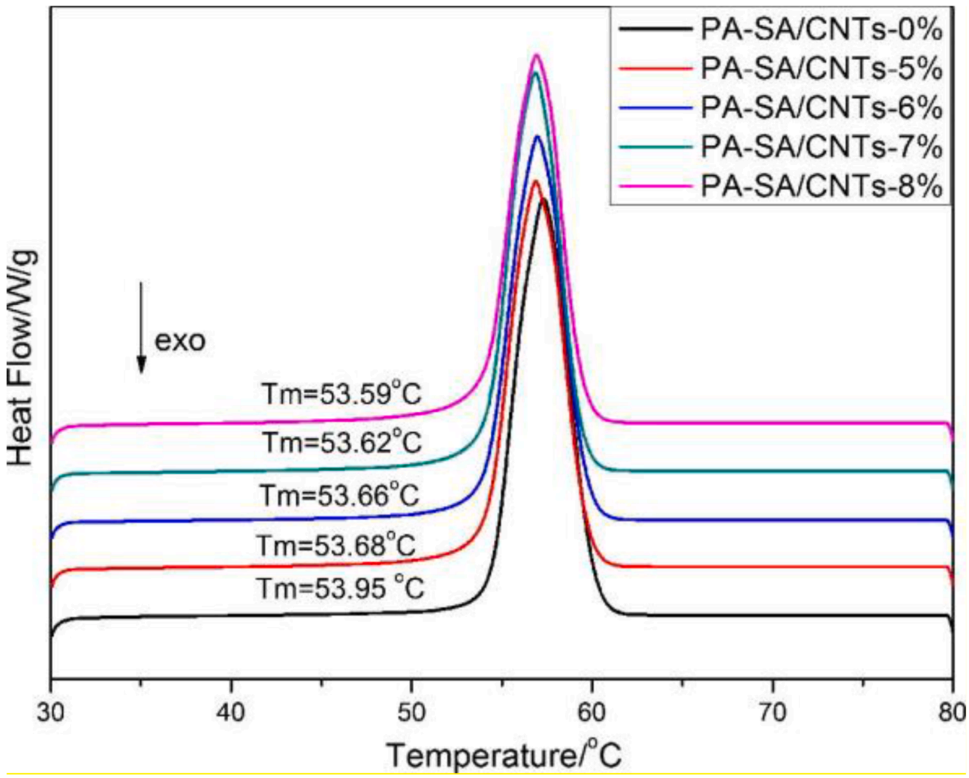

Fig.. 10. Differential scanning calorimetry (DSC) curves of PA–SA and PA–SA/CNTs composite phase change materials having different mass fractions of CNTs during the melting process[174].

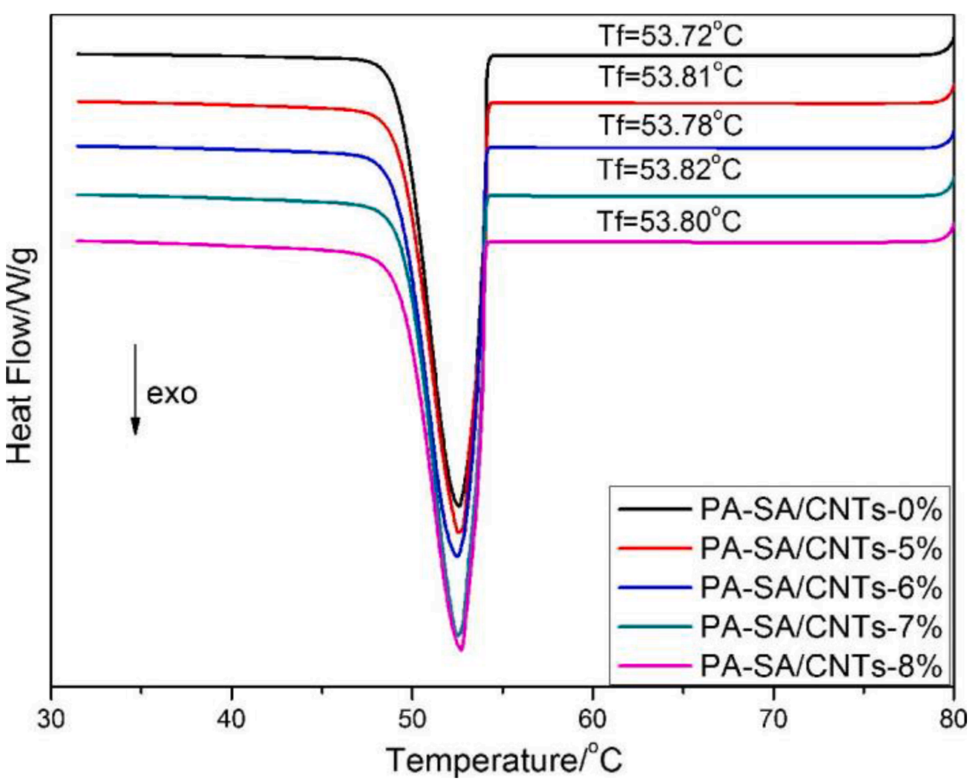

Fig.. 11. DSC curves of PA–SA and PA–SA/CNTs composite phase change materials with the different mass fraction of CNTs during the freezing process[175].

(PCMs) having carbon nanotube as an additive. The influence of the mass percentage of carbon nanotube (CNT) on the thermal conductivity as well as thermal storage/discharge rate of the composite phase change materials has been evaluated. Furthermore, the thermal stability has been additionally examined through thermal cycling analysis. The individual palmitic acid-stearic acid (PA–SA)/CNT composite phase

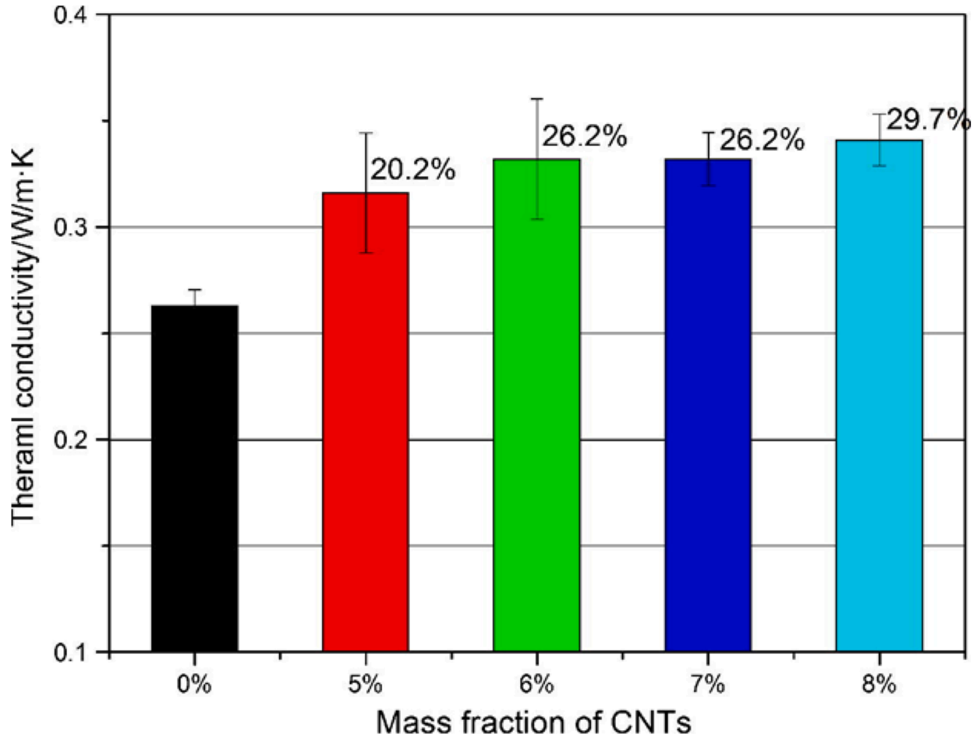

Fig.. 12. Thermal conductivity of PA–SA/CNTs composite Phase Change Materials with a different mass fraction of CNTs[175].

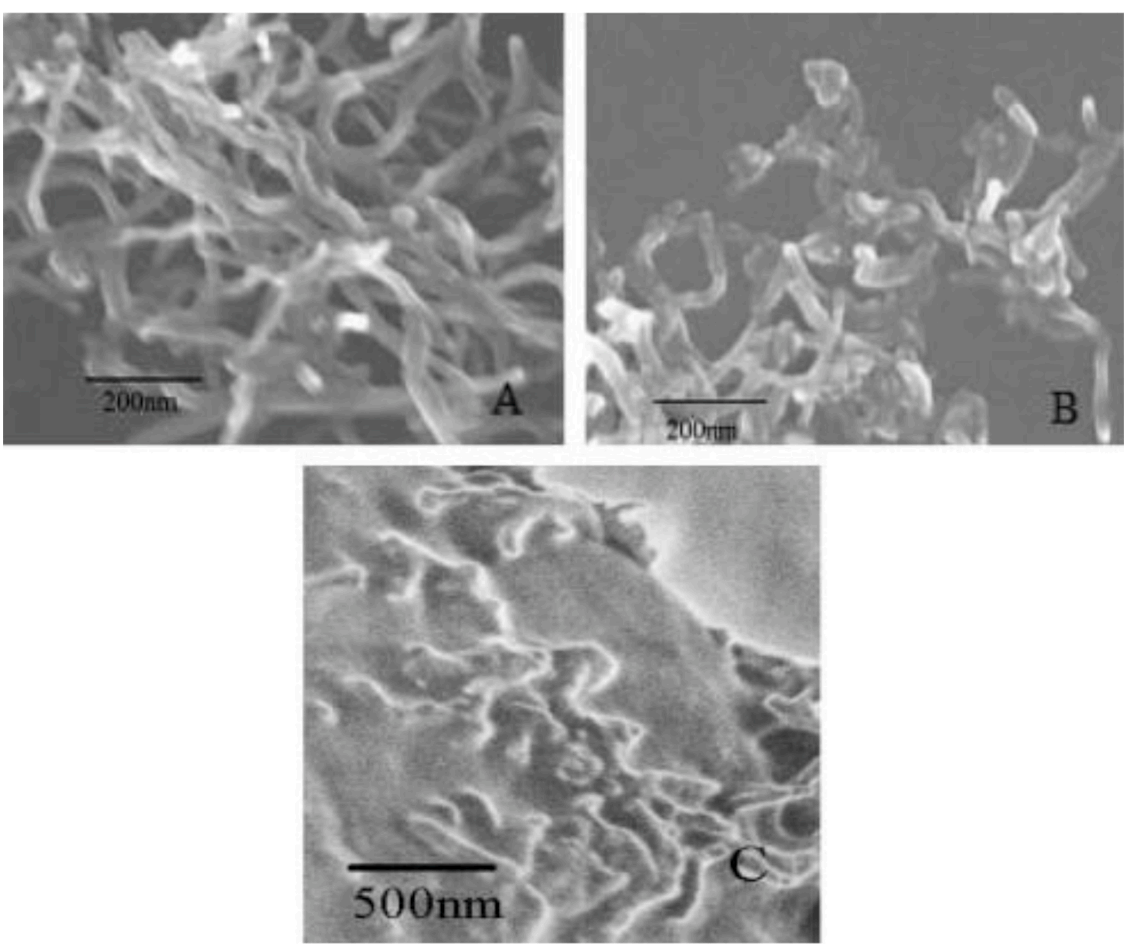

Fig.. 13. SEM images of PCNTs (A), TCNTs (B), and PA/TCNT (C)[176].

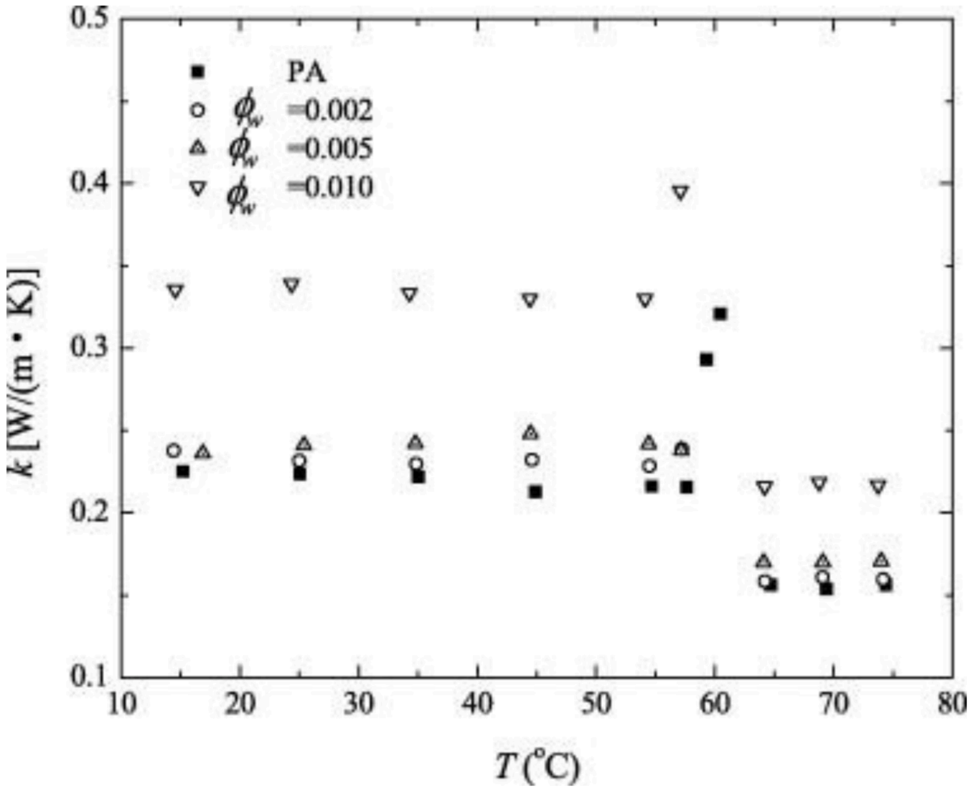

Fig.. 14. A temperature-dependant thermal conductivity of PCMs [178].

change materials reveal an appropriate thermal performance for air conditioning, solar energy storage applications. Fig. 10 illustrates the DSC curves of the Palmitic acid-stearic acid and Palmitic acid-stearic acid (PA–SA)/CNT composite phase change materials (PCMs) during the melting process. It can be observed that the DSC curves of the (PA–SA)/CNT composite phase change materials agree with that of the palmitic acid-stearic acid except that the phase change temperatures are marginally lower and present a falling trend as the higher mass fraction of carbon nanotubes(CNTs). Enhancing the mass per cent of carbon nanotubes can additionally improve the heat transfer abilities. The melting point and width at half height reduce with the raising of CNTs content. In conclusion, the PA–SA /CNT phase change materials provide tremendous thermal properties for the high melting latent heat. The latent heats of the palmitic acid-stearic acid/carbon nanotube composite phase change materials reduce with the enhancement of the amount of carbon nanotube because only palmitic acid-stearic acid retains latent heat the PA–SA /CNT composite phase change materials throughout phase change steps [174].

The DSC curves of palmitic acid-stearic acid (PA–SA), as well as palmitic acid-stearic acid/a CNTs composite phase change material throughout the freezing process, is demonstrated in Fig. 11. Due to the addition of CNTs, PA–SA/CNTs composite phase change materials offered lower melting temperatures but greater freezing temperatures contrasted to the PA–SA [174].

Fig. 12 displays the thermal conductivities of the PA–SA/CNTs composite. It is noticed that the thermal conductivity of PA–SA is enhanced by adding CNTs.The inclusion of CNTs that have been distributed and dispersed in the PA–SA, which develop a thermal conductive network. Therefore, the higher thermal conductivity of PA–SA/CNTs composite phase change materials have been achieved [175].

A unique strategy for improving the thermal performance enhanced phase change material has been reported. Fig. 13 demonstrates the scanning electron microscope images of the pristine CNT (PCNTs), treated CNTs (TCNTs) and the nanocomposites formulated with PA and TCNTs (PA/TCNTs). It has been noticed that TCNTs appear shorter in comparison with the PCNTs after mechano-chemical treatment. In the course of the mechano-chemical process, the surfaces of the CNTs is altered. The nanocomposites were obtained by adding the TCNTs into melting PA in a blending vessel. Extreme sonication has been additionally applied to create well dispersed as well as homogeneous mixtures. Fig. 15(c) shows the well-dispersed system of TCNTs/ PA matrix [176].

The connection between the thermal conductivities as well as temperatures is similarly essential for phase change materials. Fig. 14 indicates the temperature reliant thermal conductivity of PA and the PA/ TCNT composites. It has been well noticed that the thermal conductivity of the PA/TCNT composites has been improved in comparison to the PA. Besides, the thermal conductivities of the PA/TCNT composites enhance due to the increasing the mass percentage of the TCNTs (ϕw). Moreover, the thermal conductivity of PA, as well as the PA/TCNT composites, relies on the temperature. In the role of a phase change materials, high thermal conductivity near to the phase change temperature is attractive for thermal energy storage application [177]. Finally, an overview of the fatty acid /CNT phase change material is shown in Table 5.

4.1.1.5. Graphene-based fatty acid phase change material. Graphene is one layer of graphite. It possesses a perfect two-dimensional(2D) framework of carbon hexagons. The robust, as well as anisotropic sp^2 bonding along with the low mass of the carbon atoms, provide graphene exceptional thermal characteristics. At room temperature, the in-plane thermal conductivity of graphene is highest of any identified materials, about 2000−4000 W/(m K), which is greater than that of copper [190].The excellent thermal conductivity renders graphene as a desirable material for thermal management, acting in the form of thermally conductive fillers in polymer composites. Huge range of graphene nanosheets is commonly created by solution methods such as the reduction of graphene oxides or direct exfoliation of graphite for useful applications [191]. Nevertheless, the outstanding thermal

Table. 5
Properties and Application of Carbon nanotube/Fatty acid Phase change materials.

Carbon nanotube and Fatty acid based Change Material	Remarks	References
Mechanochemical reaction modified(M)- Multi-walled carbon nanotubes (MWCNTs) with palmitic acid	The thermal conductivity of the Multi-walled carbon nanotube /palmitic acid composite phase change materials is attained at the highest value by the inclusion of 1.0% Multi-walled carbon nanotube.	[179]
CNTs, oxidised carbon nanotubes and grafted carbon nanotubes /palmitic acid phase change composites	Modified carbon nanotubes exhibited advantageous latent heat and a 34.1% improvement of thermal conductivity in contrast to palmitic acid. Furthermore, the composite using modified carbon nanotube carbon nanotubes held homogeneous after 100 melting as well as freezing cycles and the retention rate of latent heat is achieved 98.5%.	[180]
Stearic acid (SA)/expanded graphite phase change materials	In comparison to pure stearic acid, the thermal conductivity of phase change composites is found to be 6.2 times greater using 9 wt% expanded graphite, The heat storage, as well as heat discharge performance, is also enhanced, appealing for low-temperature solar energy storage.	[181]
Stearic acid /carbon nanotubes phase change materials	In comparison to pure stearic acid, the thermal conductivity of phase change materials has been increased up to 1.4 times with 9 wt% carbon nanotubes, appealing for low-temperature solar energy storage.	[182]
CNT filled with lauric acid phase change materials	The thermal conductivity of the composite Carbon nanotubes / lauric acid has been found to four times greater in comparison with that of pure lauric acid, phase change materials along with much better thermal properties as a result of appropriate porous materials and modifying their pore structures.	[183]
Stearic acid and acid-treated carbon nanotubes	Phase transition of stearic acid/ acid-treated carbon nanotubes (CNTs) composite can be operated by daylight for the energy storage/discharge. Consequently, this study offers a unique platform for enhancing solar application and comprehending the phase transition characteristics of organic PCMs in dimensionally confined conditions.	[184]
CNT on thermal behaviour of the palmitic–stearic acid binary eutectic mixture	The thermal test displayed that the thermal discharge rate of the created phase change materials enhanced because of the inclusion of Carbon nanotubes although the storage rate is reduced. Thermal cycling results present that the palmitic–stearic acid binary eutectic mixture/ CNTs phase change materials held better thermal stability pure system.	[185]
Fatty acids/CNTs composite	The thermal conductivity of the composite materials has been considerably enhanced because of the inclusion of the significantly thermal conductive of the carbon nanotubes. Phase change temperatures as well as latent heat practically unchanged up to 30 thermal cycling.	[186]
Lauric acid impregnates in CNTs phase change material	Heat, as well as the mass transfer of Lauric acid/carbon nanotube phase change materials, have been improved. This study provides a profound potential for heat dissipation in nanodevices.	[187]
Multi-walled carbon nanotubes/ Lauric acid phase-change composite	The inclusion of carbon nanotubes can tune the phase change temperature as well as the latent heat of the phase change materials. The increasing the proportion of carbon nanotubes, the narrower the melting process, the higher the latent heat has been achieved. The composites offer adequate storage stability, and this composite is employed for developing energy storage materials.	[188]
Stearic acid (SA) and multi-walled carbon nanotube (MWCNT)	The multi-walled carbon nanotube(MWCNT) is consistently dispersed and distributed in the phase change material of stearic acid, and the melting point of stearic acid/ multi-walled carbon nanotube nanocomposite relocated to a lower temperature throughout the charging and then the freezing point moves to a higher temperature throughout the discharging phase in comparison to the pure stearic acid. The inclusion of multiwalled carbon nanotube increase the thermal conductivity of stearic acid successfully; also the multiwall carbon nanotube improves the heat transfer efficiency for the thermal energy storage system.	[189]

characteristics of graphene are never accomplished from graphene sheets made by chemical strategies. The remaining functional groups, as well as imperfections present on graphene sheets from chemical techniques, is unavoidable, resulting in considerable degradation of thermal features. Thermally or chemically reduced graphene sheets, as well as exfoliated graphene sheets, are usually employed to improve the thermal conductivity of various organic materials. A considerable amount of low-quality defect graphene sheets is used due to its thermal properties [192]. Previously, high-temperature thermal annealing photocatalytic and plasma reduction have been suggested as a solution to remove defect and produce high-quality graphene sheets with better electron mobility; eventhough, residual oxygen groups and defects still occur from high-temperature annealed graphene sheets. Another approach, methane plasma restoration approach, by which defects of graphene oxide can be repaired efficiently, accompanied by a simultaneous reduction process. The exclusion of defects and oxygen functional groups on graphene sheets reduces effective phonon scattering centres and enhance their thermal properties. The excellent thermal conductivity, as well as ultralightweight, produce imperfection-free graphene sheets preferably as being thermally conductive additives for phase change material, phase change material composite containing

Table. 6
An overview of the fatty acid/graphene-based phase change materials.

Graphene and Fatty acid based Change Material	Remarks	References
Laurie acid based phase and with chemically functionalized graphene nanoplatelets.	The thermal conductivity of phase change material improves by 1 vol % graphene. Excellent thermal conductivity, high aspect ratio, as well as low thermal interface resistance, renders graphene the majority appropriate nanofiller applicant to advance the thermal conductivity of phase change materials.	[202]
Lauric acid (LA) and graphene/ graphene oxide complex aerogels	The Lauric acid/graphene complex aerogel composites displayed excellent electrical conductivity, high phase-change enthalpies, better heat-charging and discharging efficiency, good cyclic stability, effective phase-change reversibility and well shape stability.	[203]
Palmitic acid/polypyrrole/ graphene nanoplatelets	Graphene nanoplatelets considerable impact on improving the thermal conductivity of the composite. By doping 1.6% of graphene nanoplatelets(GNPs), the thermal conductivity, as well as thermal capacity of palmitic acid/polypyrrole/graphene, attain up to 0.43 W/m K and 151 J/g, correspondingly. The developed phase change materials are expected to offer enhanced efficiency for solar thermal energy storage applications.	[204]
Palmitic acid/graphene -oxide phase change materials	The form-stable composite phase change materials(PCMs) posses excellent thermal stability and chemical stability. The thermal conductivity of the composite phase change material has been increased by more than three times.	[205]
Fatty acid eutectics/expanded graphite composites	It revealed excellent thermal stability after thermal treatment cycles. These types of composite are applied as potential materials for the useful radiant cooling system.	[206]
Stearic acid (SA) encapsulated by water-dispersible graphene functionalized with poly(vinyl alcohol)	The thermal conductivity of the composite is improved because of the thermally conductive graphene shell. Furthermore, the graphene shell using outstanding thermal stability as well as barrier feature efficiently acts as a protective layer for the steric acid core for enhancing thermal stability.	[207]
Palmitic-stearic acid/ graphene nanoplatelets/ expanded graphite composite	The thermal energy storage and discharge rates of the composite additionally enhanced because of the excellent thermal conductivity of carbon materials. The density of composite has been noticed to enhance with the inclusion of graphene nanoplates. All composite offer excellent thermal stability. This work illustrates that this composite has better possibilities to enhance the thermal energy storage performance of phase change material.	[208]
		[209]
Palmitic acid/ nitrogen-doped graphene composite	The thermal conductivity of the Palmitic acid / nitrogen-doped graphene composite has been established by the laser flash technique. The thermal conductivity enhanced by 500% for the 5 wt% content of nitrogen-doped graphene. Composite has excellent thermal stability and chemical reliability after 1000 melting and freezing cycles.	
Palmitic acid/high-density polyethylene/graphene nanoplatelets	The phase change materials composite offers sufficient thermal stability, and the leakage of the Palmitic acid declines considerably with the support of the graphene nanoplatelets. It is estimated that the graphene-based composite has possible use in solar energy as well as building heating systems.	[210]
Stearic acid with magnetite/ graphene nanocomposite	Fe_3O_4/Graphene offers excellent thermal stability to the stearic acid. The inclusion of the Fe_3O_4/ Graphene has improved sensible heat and the latent heat of the stearic acid.	[211]
Graphene-decorated silica stabilised stearic acid composite	The thermal conductivity of the composite has been improved by the addition of graphene. Composites have excellent thermal stability and ideal for thermal energy storage.	[212]
Lauric acid(LA) phase change nanocomposite with graphene nanoplatelets(GNPs)	Graphene improved the thermal conductivity of the lauric acid. Composite show similar phase change enthalpy as well as melting temperature in comparison to the pristine material.	[213]

defect-free graphene sheets showed a significantly enhanced thermal conductivity and lowered reduction of the heat of fusion [193]. Generating an excellent thermal transport system using graphene as nanofillers at a little volume fraction can allow transformational efficiency in dealing with the essential technological difficulty to build advanced PCMs along with high thermal transfer as well as energy-storage abilities [194]. Also, the thermal conductivity of graphene nanosheets is substantially better thermal conductivity of CNTs [195]. Therefore, it is expected that the application of graphene can considerably enhance the thermal conductivity of organic fatty acids based on phase change materials. Because, the direct application of fatty acids for thermal energy storage displays a few challenges due to their inferior odour (as a result of sublimation during heating), poor thermal conductivity, and inadequate thermal stability [196]. As an outcome, shape-stabilised composite phase change materials have been developed to fix these challenges by using the full capabilities of organic thermal energy storage and high thermal conductivity properties of organic or inorganic materials [197]. Authors have applied graphene nanosheets, and they have mentioned that reported a thermal conductivity of fatty acid-based ester nanocomposites have been enhanced by loading of 1 vol% of graphene nanosheets [198]. Authors have developed lauric acid based phase change nanocomposites using chemically functionalized graphene nanoplatelets (GNPs). Introduction of graphene nanoplatelets (GNPs) improves the thermal conductivity of phase change material (PCM). The phase change enthalpy, as well as the melting temperature, is still identical in compare with pristine material, that renders this material as an appealing applicant for thermal energy storage applications [199]. Self-assembly strategy is discussed to create a phase change material

Table. 7
Application of fatty acid and carbon allotropes based phase-change materials.

Fatty acid and Carbon Allotropes based Change Materials	Remarks	References
Organic PCM based on lauric acid (LA) from coconut oil (CO)	The air temperature decreases which helps to use this material for the Air Conditioning (AC) system and human's thermal comfort.	[220]
Capric acid (CA)-myristic acid (MA)/ cement composite	Long-term cycling reliability and chemical stability, application in low-temperature ventilating and air conditioning intentions in buildings.	[221]
Silica fume/capric acid-palmitic acid/ CNT composite	Improvement in thermal conductivity by CNTs doping, good cycling thermal reliability and chemical stability, application in passive solar thermoregulation of building envelopes.	[222]
Stearic acid and hexanamide eutectic mixture	Excellent thermal stability and reliability, application in thermal energy storage	[223]
1-Hexadecanol-Palmitic Acid Eutectic Mixture/Activated Carbon Composite	Excellent durability, thermal stability and thermal energy storage-release rate, application in energy-saving building materials.	[224]
Stearic acid-benzamide eutectic mixture/expanded graphite composites	Excellent thermal and chemical stabilities, great potential in thermal energy storage application.	[225]
Ultrathin graphite sheets stabilised stearic acid composite	Remarkable latent heat for thermal energy storage applications involving cooling, building energy efficiency.	[226]
Myristic acid (MA)-palmitic eutectic mixtures encapsulated in PMMA shell	Excellent thermal/chemical stability, application in thermal energy storage (TES) systems for building components.	[227]
Capric-stearic acid/White Carbon Black composite	High thermal storage density, suitable melting temperature and thermal conductivity, outstanding thermal stability an reliability, application in building envelope.	[228]
Microencapsulated myristic acid with titanium dioxide (TiO2) shell	Excellent thermal reliability after 1000 repeated melting/freezing cycling. Application in building components.	[229]
Capric Acid/Cetyl Alcohol Binary Eutectic Phase Change Material	Excellent thermal stability over 1000 thermal cycles, application in cold thermal energy storage.	[230]
Fatty acids based eutectic phase change system	Superior thermo-physical characteristics with no supercooling, thermal energy storage ranging from the room temperature to 75 °C for building	[231]
Fatty acid/diatomite form-stable composite	Outstanding thermal properties, good chemical and thermal reliability and high thermal conductivity, application in modern buildings	[232]
Octanoic-lauric acid/expanded graphite	Good stability after cycle testing, application in the medical refrigeration transportation system and air conditioning cold storage system with a temperature range of 2–8 °C.	[233]

(PCM) composite that consists of palmitic acid (PA) as well as oleylamine modified-reduced graphene oxide as an assisting material. This method removes typical freeze-drying and also impregnation stages. Also, there is no shrinkage throughout the drying procedure. Graphene network provides crystallisation of palmitic acid with better heat transfer, a terrific shape-stable feature to phase change material, which avoids the leakage of molten palmitic acid [200]. A unique shape-stabilised phase change material of stearic acid–graphene oxide has been designed by the formation of the core-shell structure of the stearic acid and graphene. This Phase change material in nanoconfined geometries has been explained. The thermal properties of stearic acid–graphene oxide composite are structured by changing stearic acid/graphene oxide mass ratio. The stearic acid – graphene oxide composite developed from stearic acid/graphene oxide mass ratio of one displayed significantly minimized melting point (Tm) as well as freezing point (Tf) in comparison to pristine stearic acid, and the latent heat of fusion was found 55.7 J/g complimenting to the thermal storage capability rate of 82.4% [201]. Core-shell stearic acid /graphene microcapsule has been developed by a latex technology. Graphene encapsulated stearic acid latex particle in aqueous medium has been achieved through electrostatic attraction, which helps to develop a core-shell composite microcapsule. The graphene shell provides outstanding shape stability to the composite throughout phase change. Also, the core-shell structure consists of the ultra-thin graphene shell, which has elevated thermal conductivity. This composite has significant latent heat and enhanced thermal conductivity. There are many studies, which provide the impact of graphene on the phase change behaviour of a fatty acid; we have reviewed some of the important studies, as mentioned in Table 6.

5. Application of fatty acid and carbon allotropes based phase change materials

The application of phase change material(PCM) has grown progressively in various industries, such as the electronic industry, solar dryers in the agricultural industry, waste heat recovery systems, solar cooling and solar power plants, photovoltaic electricity systems, pharmaceutical products and preservation of food, domestic hot water and space industry. Aside from the other applications, thermal comfort and energy performance in buildings are being improved by phase change materials. Therefore, phase change material applications could be a commanding tool in net-zero energy buildings architecture. The content of phase change material depends on the specific thermal storage application. As, in the case of building integrated latent heat storage, phase change material can be contained in a porous matrix (concrete, wood, plasterboard, etc.). However, various shortcomings of phase change material, such as leakage during the melting process,the low thermal conductivity, and weak receptiveness to optical or electrical stimuli hinder their practical applications. To overcome these shortcomings, scientists have tried to implant various phase change material into a lot of matrices, such as microcapsules and 3D porous materials (sponge, aerogel and foam) with an interconnected network. The matrices can provide the phase change material better thermal conductivity, extreme absorption of sunlight or sufficient electrical conductivity, shape stability during the melting process for multiple driving ways. In principle, the ideal matrix should possess high mechanical property and large capillary force to avoid leakage during working, outstanding thermal conductivity to increase the power capacity, frivolous framework to maintain high latent heat, and high electrical conductivity and dark colour to store thermal energy via multi-driven ways (sunlight absorbing or Joule heating). It is beneficial that in order to enhance the influence of the phase change materials(PCMs), it is essential to select a phase change materials with an appropriate phase transition temperature. Phase change materials in solar energy systems, recovery heat systems have been primarily employed for their large latent heat storage capacity, which is employed to modify the imbalance between heat supply as well as demand. Phase change materials(PCMs) in buildings, textiles and cooling systems have been primarily applied to manage an optimal environmental temperature by constant phase change temperature.

Construction and structure industry is a leading customer of material as well as energy resources that address nearly 40% of consumption. However, confined traditional fossil energy sources build dangerous

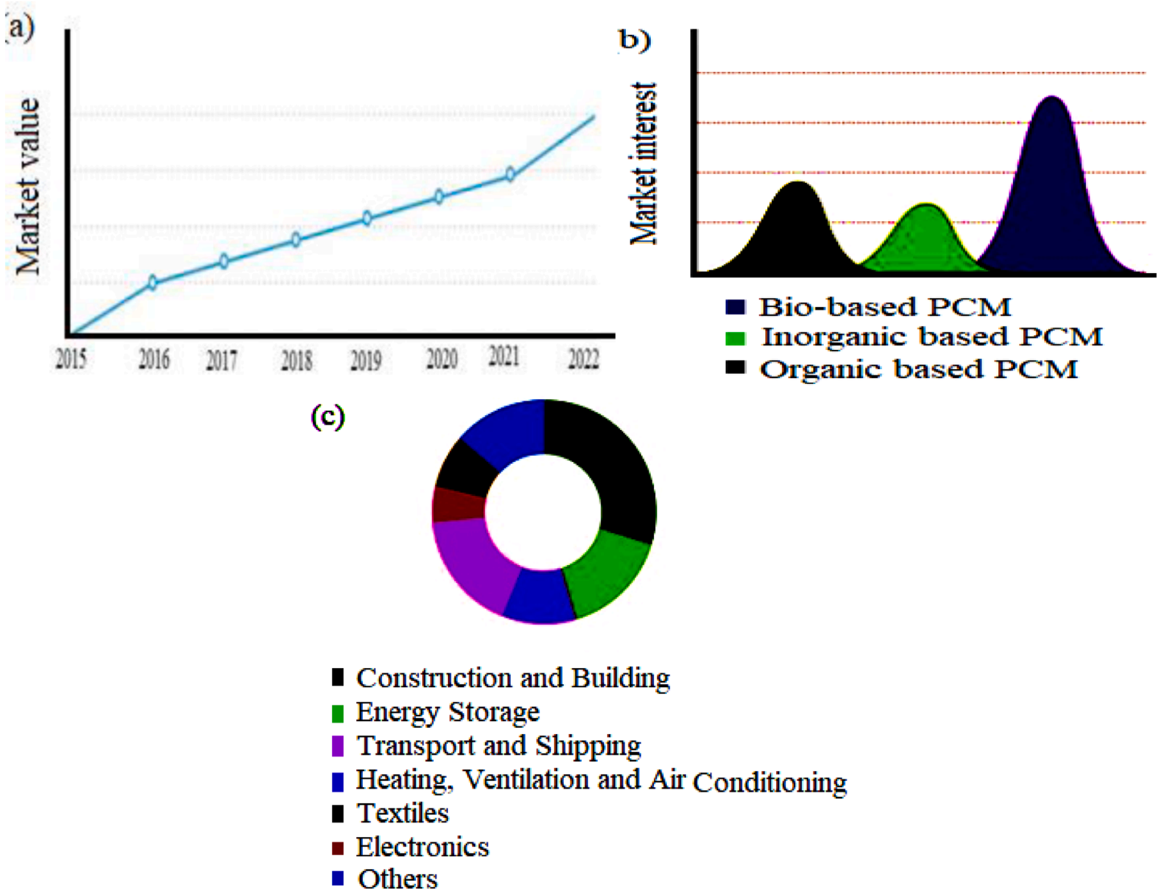

Fig.. 15. (a) Global phase change material market, (b) Market of various types of phase change material globally, (c) Application of Phase change materials(From Google database).

emissions that are responsible for environmental pollution. Practical as well as affordable technology which can be applied to store significant levels of heat or cold in a specific volume is the topic of study for a long time. Thermal storage performs a crucial part in construction energy conservation, that is considerably improved by the inclusion of latent heat storage materials in building items. In an attempt to preserve energy, thermal energy storage systems are considered an appropriate solution [214]. Thermal energy storage can save energy for future usage by application of latent heat storage or sensible heat storage materials. Present thermal energy storage materials have been applied in the construction industry consists of sensible heat storage materials such as steel, water, wherein thermal energy is accumulated by lifting the temperature of the material. With the continuous enhancement of living requirements, the requirements for thermal comfort indoors buildings is boosting, resultant additional energy consumption, particularly in the winter and summer season [215]. Establishing phase change material into buildings reduces energy consumption and magnifies the thermal comfort of buildings complexes [216]. Phase change materials tend to be able of storing energy at a particular or steady temperature that is called the phase transition temperature of the phase change materials(PCMs). The passive and active systems based on PCM are generally used in building in applications. The collecting, storing, and releasing heat by building structure itself is executed by a passive system. In addition to the active system, pumps or fans is used to provide a heat transfer medium. Phase change material(PCM) can be used the building as building walls, floor, ceiling, building interior. The main advantage of PCM in the building are: 1) it can improve the human thermal comfort; 2) PCM based wall slowdown the overheating effect; 3) Heat transfer phenomena such as convection is also improved by using PCM, 4) The phase change has a stabilizing effect on the inner surface temperature of the ceiling [217]. For example, incorporation of phase change materials into construction fabric improves the thermal storage capacity of the building. The experimental outcomes presented that phase change materials based panels can diminish the temperature fluctuation of the room, phase change materials based panels also offer a smaller range of temperature in comparison with the traditional room. Additionally, the thermal efficiency of the room is linked to the location of phase change materials based panels [218]. Integration of phase change Materials in concrete and wallboards has regained interest in the past few years. Wallboard is looked appropriate for incorporating the phase change materials because wallboard has several advantages such as larger heat exchange area, lightweight, inexpensive. The performance PCM-wallboard depends on the types of phase change materials, processing method, the surrounding condition, the rate of ventilation. The microencapsulated phase change material has also been applied in building in the form of an insulation system. The main advantage insulation system based on the phase change material is reducing and shifting the peak hour thermal load of the building. However, design an insulation system based on the phase change material is a crucial process. There are several research, which has been focused on the use of phase change material in building components such as wall core, containers, wall core, pipes, wall, gypsum board, roof, roof desk, ceiling. The floor performs the heating and cooling of a building, with constant temperature, high energy storage density is received by PCM incorporated floor, PCM is charged at night due to low electric demand, and discharged in a day. The heat loss in building mainly take place by the window, PCM based shutter help to decrease the heat loss from the window [219]. An overview of Fatty acid based phase change materials in building the application is mentioned in Table 7.

6. Current market of phase change materials

The phase change materials(PCMs) market produced earnings of $691 million in 2015, which is supposed to achieve $ 2, 434 million by 2022. The worldwide phase change materials industry is divided based on form, practical application [234]. Construction & building section is expected to be the leading investment area in the worldwide for phase change materials market, due to the considerable use of these types of materials in construction interior design and development of the structure industry in China, India, as well as Brazil. Furthermore, interest in "green" construction is anticipated to devote to the progress of the current market, as shown in Fig. 15.

7. Conclusions and future prospects

The industrialisation of the latent heat storage unit is seen to be confined because of insufficient suitable thermo-physical characteristics. The low thermal conductivity of phase change material, change in thermo-physical capabilities of phase change materials within prolonged cycles, phase segregation, subcooling, incongruent melting, volume alteration. The main disadvantage is the poor thermal conductivity as well as the thermal stability of the phase change material. The influence of the lower conductivity value is shown in the course of energy retrieval or even elimination with a significant temperature reduction throughout the process. Because of this, the rate of phase change process (melting/solidification of phase change material) has not been reached the predicted intensity. To put it briefly, a sufficient level of energy can be accessible. However, the system is probably not able to utilise it at the mandatory rate. Additional, the choice of using phase change material in almost any latent energy storage system is determined by suitable kinetic, thermo-physical, and also chemical attributes in combination with financial requirements. However, every category of phase change material possesses its own features, functions, positive aspects, and also boundaries. Since no single material, The cost-effective criterion for using a phase change material in a system is dependant upon the lifespan price of the storage material. Therefore, to assure successful functionality and also financial feasibility of latent heat storage systems an extensive understanding of thermal stability of the phase change materials just as features of the repeated range of thermal cycles is essential.

Boosting the heat transfer performance of phase change material is a critical issue in both energy and economic parts. Numerous techniques can be utilised to boost the poor heat transfer of phase change materials. Including using extended surfaces, the mostly adopted approaches employing multiple phase change material units, heightening thermal conductivity and formulating micro-encapsulated of phase change material. amidst these technologies, using extended surfaces is the most upfront approach. The adaptable method is increasing the thermal conductivities of phase change materials, and the corresponding main methods include dispersing high thermal conductivity particles, both in micro-size and nanosize into phase change material, infiltration of phase change material into the porous structure with high thermal conductivity and high porosity. In conclusion, phase change materials have an outstanding role in protecting the environment as well as saving energy. They are essential materials in the future development of energy advancement.

Declaration of Competing Interest

None

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